CV22FS55 Datasheet - Preliminary

SUMMARY DESCRIPTION

The Ambarella CV22FS55 SoC combines the CVflow® computer vision / deep learning accelerator, an industry-leading image signal processor (ISP), and a H.264 video encoder to enable a variety of low-power, high-resolution automotive sensing and viewing camera systems.

Target applications include single- and multi-camera advanced driver assistance systems (ADAS), driver monitoring systems (DMS) and in-cabin solutions, single- and dual-channel electronic mirrors, and intelligent parking assistance systems.

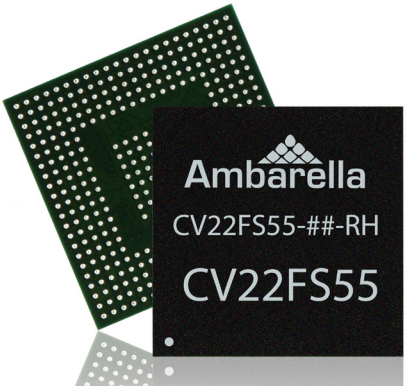
The CV22FS55 chip is fabricated using low power (LP) 10 nm silicon technology, minimizing system power requirements.

KEY FEATURES

- An embedded quad-core Arm® Cortex®-A53 1 GHz CPU with 1 MB L2 cache
- Arm Cortex-R52 456 MHz CPU with DCLS (dual-core lockstep) with NEON support
- Industry-leading ISP with HDR processing, LED flicker mitigation, dewarping, and 3D motion-compensated temporal filtering (MCTF)
- High-resolution video codec (H.264) to enable video recording and streaming applications
- CVflow vector processor with convolutional neural network (CNN) / deep learning support
- Dense optical flow accelerator
- ISO 26262 compliant to meet ASIL C
- Advanced suite of hardware cybersecurity features for secure boot, software over-the-air update (OTA), and secure communication
- 441-pin, 0.65-mm pitch FC TFBGA package (14 mm x 14 mm)
- AEC-Q100 Grade-2
- Operating junction temperature (T_J): -40 C to +125 C

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1. OVERVIEW

This preliminary datasheet for the CV22FS55 SoC from Ambarella begins with a brief introduction to the chip (Section 1.1) and a summary of key features (Section 1.2). Chapter 2 describes the CV22FS55 peripheral interfaces. For pin details and electrical characteristics refer to Chapter 3 and Chapter 4, respectively. See Chapter 5 for package information, and Chapter 6 for Ambarella contact and ordering details.

Note that the chip features described in this datasheet are subject to change. Details that have not been entirely finalized (e.g., encoding specifics) are provided using conservative estimates (i.e., final encoding performance is expected to meet or exceed the estimate provided). Contact an Ambarella representative for additional information.

1.1 Introduction

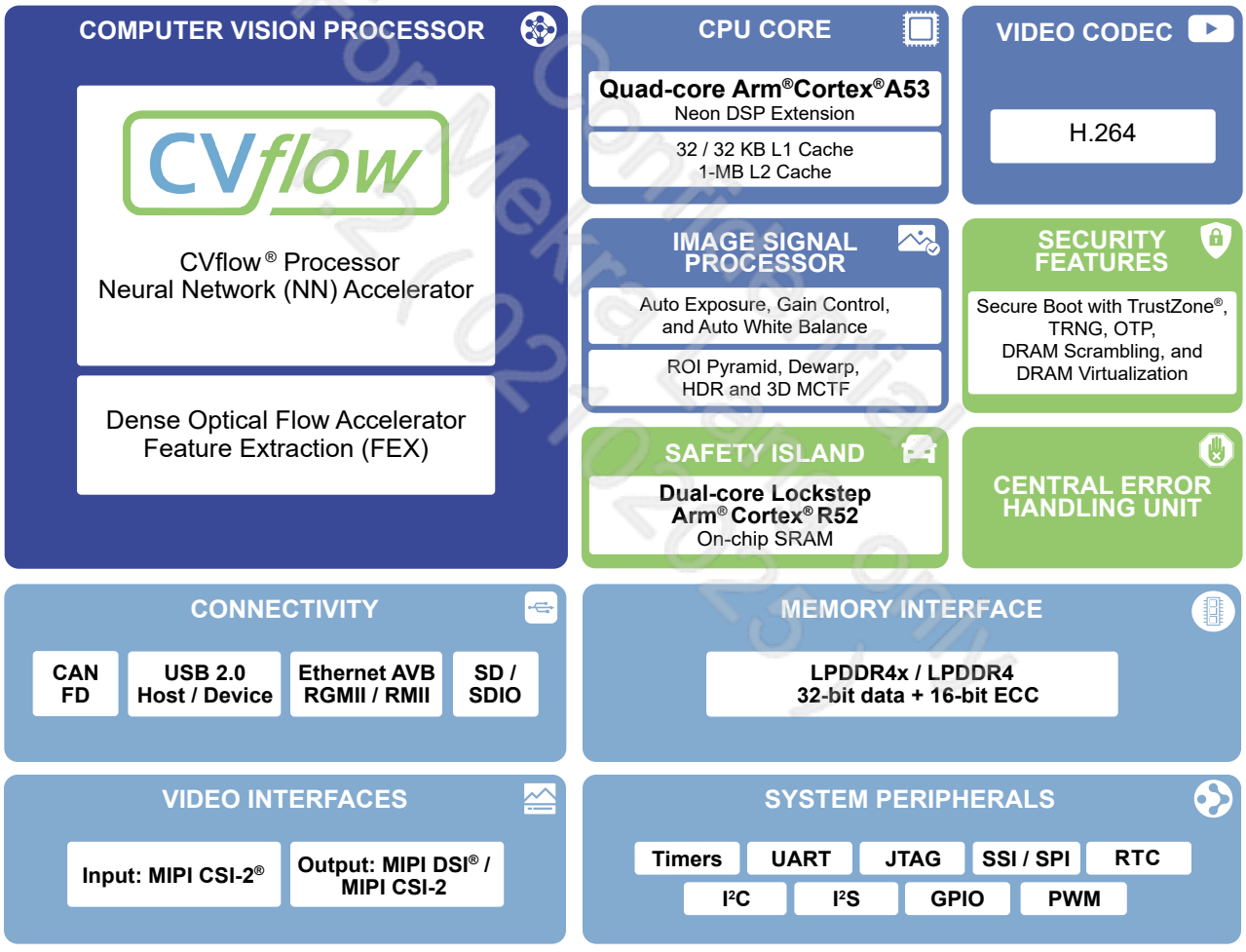


Figure 1-1. CV22FS55 Overview: Functional Block Diagram of the CV22FS55 SoC.

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1.2 Feature List

Features of the CV22FS55 chip include the following:

1.2.1 Embedded Quad-Core Arm Cortex-A53 CPU

- Clock frequency up to 1 GHz
- 32 KB data / 32 KB instruction cache
- 1 MB L2 cache
- NEON SIMD acceleration

1.2.2 Arm Cortex-R52 CPU with DCLS (Dual-Core Lockstep) with NEON Support

- Clock frequency up to 456 MHz
- 32 KB L1 cache
- 1 MB ATCM

1.2.3 LPDDR4x / LPDDR4 Controller

- Up to 1.8 GHz clock rate
- 32-bit data bus for data and 16-bit data bus for ECC
- Up to 2 GB capacity

1.2.4 Computer Vision Processing

- CVflow processor
  - Low-latency, high-throughput computer vision processing
  - Acceleration for neural network convolution / fully-connected layers
  - Up to 4D vector at 8-bit fixed point, 16-bit fixed point, and 16-bit floating point
  - Arithmetic and logical element-wise processing
  - Affine transform
  - Resampling
  - Gather / scatter operators
  - Interpolation (up to 3D)
  - Transposition / reshape
  - Min / max reduction
  - Vector statistics
- Dense optical flow accelerator
  - 8-bit YUV input
  - Sub-pixel accuracy of matching vectors
  - Flow vector horizontal displacement range: [-16.0, 16.0)

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- Flow vector vertical displacement range: [-8.0, 8.0)
- Output best matching score and confidence value (optional)
- Image resolution up to 640w * 2032h
- Multi-scale image pyramid refinement

1.2.5 Functional Safety

- ISO 26262 compliant to meet ASIL C
  - Widespread parity or error correcting code (ECC) protection of on-chip memory and interconnect
  - Power-on logic built-in self-test (BIST) for safety logic
  - Central error handling unit (CEHU) with error pins
  - Runtime safety diagnostics, voltage monitors, clock monitoring, watchdog timers, and ECC engine for memory integrity checks
  - Safety documentation for applications required to meet ISO 26262 requirements
  - Arm Cortex-R52 core with dual-core lockstep operation via an integrated MCU subsystem
    - Up to 1 MB tightly-coupled memory (TCM) RAM with ECC dedicated for Arm Cortex-R52
  - Capable of being isolated by the external MCU

1.2.6 Sensor / Video Input (VIN) Interfaces

- Support for popular CMOS sensors inlcuding Sony, onsemi, and Omnivision
- MIPI CSI-2 (with virtual channel)
- Multi-sensor support
- The video interfaces can support Bayer RGB (RAW) or YUV422 video formats
  - 8- / 10- / 12- / 16-bit Bayer RGB (RAW) formats

1.2.7 Image / Video Processing

- Up to 300 MPixel/s
- RGGB / RCCC / RCCB / RGB-IR(4x4) / monochrome sensor support
- High dynamic range (HDR) processing
  - Multi-exposure HDR (DOL, dual conversion gain)
  - Sensor compressed HDR (up to 16 bits)
  - 120 dB dynamic range processing with local tone mapping and contrast enhancement
- Motion-compensated temporal filtering (3D noise reduction) for excellent low-light performance
- Geometric lens distortion correction
- LED flicker compensation for LED headlights, traffic signs, and other LED sources in the scene
- Chromatic aberration correction
- Vignetting compensation
- Programmable multi-FOV image scaling and cropping
- Black-level correction

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- Static and dynamic bad pixel correction
- Auto exposure / gain control and auto white balance (AE and AWB)
- Digital gain
- Edge aware adaptive sharpening

1.2.8 Video Output (VOUT) Interfaces

- CV22FS55 drives two video output data types simultaneously
  - Two four-lane MIPI DSI® / MIPI CSI-2® outputs
- Flexible OSD with 8- , 16- , and 32-bit options
- CV22FS55 provides flexible graphics and OSD capability, controlled by Arm software. Graphical overlays can be implemented on recorded video streams, video output to display, or both.

1.2.9 Video Encoding

- Broadcast-quality H.264 Main Profile level 5.1 video encoding and decoding
- Encoding / decoding performance up to 240 MPixel/s (8MP30)
  - Full duplex (simultaneous encode / decode) mode support
  - Flexible encode resolution control including aspect ratio, scan type, and frame rate
- Multi-channel / multi-stream encoding
- Picture in picture (PIP) or picture in video (PIV)
- Interval (or time lapse) video encoding with live preview
- Video encode pause / resume
- Video with embedded thumbnails (configurable size and number)
- Flexible bitrate control
- Date / time stamp

1.2.10 AHB Bus Direct Memory Access (DMA) Controller

- Two DMA controllers with each controller supporting 8 channels
- Memory-to-memory transfers including transfers between memory and peripherals
- Programmable transfer count up to 4 MB
- DMA scatter / gather via chained descriptor list in memory with DMA control information source

1.2.11 Dedicated DMA Coprocessor for Graphics and Image Operations

- Linear copy, 2D copy, composite, and alpha-blend image operations
- 4- to 32-bit pixel formats

1.2.12 Audio Input / Output

- One I2S controller that is capable of capturing and outputting I2S data
- The audio captured over I2S can be further pre-processed by implementing algorithms on the Arm core

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- The SDK supports a variety of different audio codecs for audio compression, which are implemented on the Arm

1.2.13 Ethernet MAC Controller

- Two ENET ports with 10 / 100 / 1000-Mbps data transfer rates with an IEEE 802.3-compliant RGMII interface
- IEEE 802.3 compliant with full- and half-duplex (IEEE 802.3x flow-control) and jumbo frames
- IEEE 802.1Q VLAN tag detection
- Audio video bridging (AVB) endpoint implementation for IEEE 1588/802.1AS with hardware timestamp
- Checksum off-load for received IP and TCP / UDP packets
- Dedicated pins for RGMII interface
- FIFO (4 KB / 4 KB) and DMA support

1.2.14 USB 2.0 Interface

- One port configurable as host or device, with built-in PHY

1.2.15 Flexible Storage Media Input / Output (SMIO) Interface

- NAND flash controller
  - Up to 128 Gb device, 2 KB, and 4 KB page sizes
  - SPI compatible serial bus
  - 4-bit and 8-bit SLC with ECC hardware and read-confirm support
  - BCH error correction and increased spare area available
- Two SD controllers (SD0, SDIO0)
  - SD0:
    - SDIO v3.0, SD, SDHC, SDXC, MMC, and eMMC operation with boot support and UHS-I speed
    - 1-bit, 4-bit, and 8-bit SD mode
  - SDIO0:
    - SDIO v3.0, SD, SDHC, SDXC, MMC, and eMMC operation with UHS-I speed
    - 1-bit and 4-bit SD mode
  - 32 GB maximum capacity for SDHC SD card
  - 1 TB maximum capacity for SDXC SD card
  - CRC7 for command and CRC16 for data integrity

1.2.16 Advanced Security Features

- AES / SHA1 / SHA2-256 crypto acceleration using Arm V8 extensions
- Secure boot
- On-chip true random number generator (TRNG)
- One-time programmable (OTP)

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- Arm TrustZone® with secure memory
- DRAM scrambling
- DRAM virtualization
- Programmable secure level for each peripheral interface
- eFuse disable for JTAG and USB

1.2.17 Multiple Boot Options

- SPI NAND, SPI NOR, USB, and eMMC
- Single- / dual- / quad- / octal-SPI NOR
- Single- / dual- / quad-SPI NAND

1.2.18 Generic Interrupt Controller including GIC CPU-Offload Functionality

- Embedded GIC inside Cortex-R52
- One external GIC to support both Cortex-A53 and Cortex-R52

1.2.19 SSI / SPI Controller Interfaces

- Nine SSI / SPI interfaces comprising of six SSI masters, one SSI slave, one SPI NOR, and one SPI NAND

1.2.20 Two-Wire Serial Inter-Integrated Circuit (I2C / IDC) Master Interfaces (x6)

- Configurable IDC buses

1.2.21 UART Interfaces (x5)

- 4 AHB-bus UART interfaces, 1 APB-bus UART interface
  - DMA support included in the 4 AHB interfaces with flow control

1.2.22 General Purpose Input / Output (GPIO) Pins

- Up to 103 pins with individual pull-up / down control

1.2.23 Controller Area Network with Flexible Data Rate (CAN FD) Controllers

- Up to 6 instances

1.2.24 Real Time Clock (RTC)

- 32.768 kHz reference clock at individual power domain

1.2.25 Timers

- Three timers instantiated in the APB bus

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- Each instance supports 10 counters

1.2.26 Watchdog Timer

- Six APB watchdog timers

1.2.27 Pulse Width Modulators (PWM) (x12)

- Three sets of PWM controllers and each controller supports up to 4 PWMs

1.2.28 Debug Interface

- JTAG in-circuit emulator (ICE) interface for debugging

1.2.29 Non-Functional Features

- 441-pin, 0.65-mm pitch FC TFBGA package (14 mm x 14 mm)
- 10 nm CMOS low power (LP) technology
- AEC-Q100 Grade-2
- Operating junction temperature (T_J) from -40 C to +125 C

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2. INTERFACES

2.1 Interfaces: Overview

This chapter summarizes the features of the interfaces for the CV22FS55 chip and includes the following sections:

- (Section 2.2) Interfaces: SDRAM
- (Section 2.3) Interfaces: Video Input
- (Section 2.4) Interfaces: Video Output
- (Section 2.5) Interfaces: Audio Input and Output
- (Section 2.6) Interfaces: Gigabit Ethernet
- (Section 2.7) Interfaces: USB
- (Section 2.8) Interfaces: SD / SDIO
- (Section 2.9) Interfaces: SSI / SPI
- (Section 2.10) Interfaces: I2C / IDC
- (Section 2.11) Interfaces: UART
- (Section 2.12) Interfaces: General Purpose Input / Output (GPIO)
- (Section 2.13) Interfaces: CAN FD
- (Section 2.14) Interfaces: Real Time Clock (RTC)
- (Section 2.15) Interfaces: Pulse Width Modulator (PWM)
- (Section 2.16) Interfaces: JTAG

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2.2 Interfaces: SDRAM

The CV22FS55 chip supports LPDDR4x / LPDDR4 SDRAM interfaces. Contact an Ambarella representative for details on programming and calibration.

Features include:

- Maximum clock frequency of 1.8 GHz
- Programmable IO strength
- 32-bit data bus for data and 16-bit data bus for ECC

2.3 Interfaces: Video Input

The features of the CV22FS55 VIN module include:

- Six video input (VIN) modules: VIN0 (main), VIN1, VIN2, VIN3, VIN4, VIN8
- The six VIN modules can be connected to video interfaces such as MIPI CSI-2
- MIPI CSI-2
  - Three 4 lane MIPI CSI-2 ports (up to 2.5 Gbps / lane)
    - Each 4 lane MIPI port can support 1, 2, 3, 4 lane MIPI CSI-2 input
    - Two 4 lane MIPI ports can be combined to support an 8-lane MIPI CSI-2 input
  - Only one out of the three MIPI CSI-2 ports supports MIPI virtual channel (lanes 1-4) (refer to

Section 3.2.3 for details)

- MIPI configurations:
  - Three (1-4 lane) MIPI CSI-2 ports connected to VIN0, VIN1, VIN8
    - One of the three MIPI CSI-2 ports can support up to four virtual MIPI channels. In this case channel 0 is captured by VIN0, channel 1 to VIN2, channel 2 to VIN3, channel 3 to VIN4. The other two ports would be connected to VIN1 and VIN8 respectively
  - One (8 lane) MIPI CSI-2 port is connected to VIN0 and one (1-4 lane) MIPI CSI-2 is connected to VIN8

Contact an Ambarella representative for information regarding VIN module configuration and other possible lane sharing combinations.

2.4 Interfaces: Video Output

The CV22FS55 video output (VOUT) has two independent video output channels.

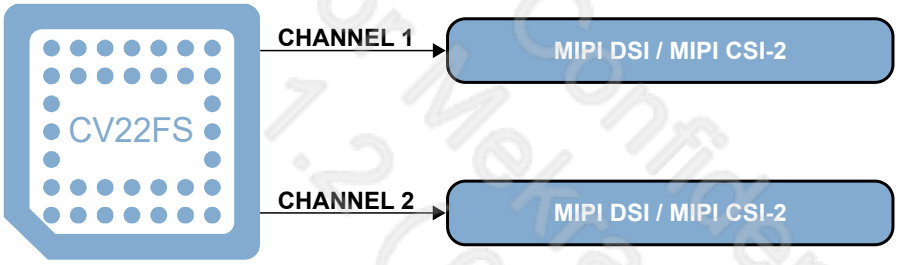


Figure 2-1. CV22FS55 Video Output Channels and Ports.

Both VOUT channels are capable for driving MIPI DSI / MIPI CSI-2 outputs. The two channels are independent and can drive different output formats.

2.4.1 MIPI DSI

The CV22FS55 chip includes MIPI DSI transmitter, supporting 1 / 2 / 4 lane MIPI output. The chip supports data transmission in high-speed mode. Command transmission is supported in high-speed as well as low-power modes.

The DSI formats that are supported are as follows:

- 16-bits per pixel, YCbCr 4:2:2
- 16-bits per pixel, RGB 565
- 24-bits per pixel, RGB 888
- 18-bits per pixel (loosely packed), RGB 666

Note:
The MIPI DSI interface can be clocked up to 2 Gbps.

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2.4.2 MIPI CSI-2

The CV22FS55 chip includes MIPI CSI-2 transmitter, supporting 1 / 2 / 4 lane MIPI output. The chip supports data transmission in high-speed mode.

The CSI-2 formats that are supported are as follows:

- 16-bits per pixel, YUV422-8bit
- 8-bits per pixel (RAW8)

Note:
The MIPI CSI-2 interface can be clocked up to 2 Gbps.

2.5 Interfaces: Audio Input and Output

The CV22FS55 chip includes one two-channel I2S controller.

- The two-channel I2S controller supports one I2S input data lane: I2S_SI_0; and one I2S output data lane: I2S_SO_0
- The controller has its own common word / LR clock and bit clock for the I2S data lanes. The data lanes are clocked by a common clock signal for the controller.
- The controller has an independent master clock to drive an external A/D converter
- The I2S clocks can be configured to operate in master / slave configuration
- The sampling frequencies supported by the I2S controller for input and output are 8 kHz, 11.025 kHz, 12 kHz, 16 kHz, 22.05 kHz, 24 kHz, 32 kHz, 44.1 kHz, 48 kHz, and 96 kHz
- Each I2S data lane typically supports a stereo channel. However, the I2S input data lines can also be configured to operate in time division multiplexing (TDM) mode, enabling multiple channels of audio to be operated on a single data line.
- Audio encode / decode including G.711, G726, ADPCM, and AAC are implemented in on-chip software

2.6 Interfaces: Gigabit Ethernet

- Two ENET ports with 10 / 100 / 1000-Mbps data transfer rates with an IEEE 802.3- compliant RGMII or RMII interface
- External Gigabit / fast Ethernet PHY
- MDIO master interface (optional) for PHY device configuration and management
- Audio video bridging (AVB) endpoint implementation for IEEE 1588/802.1AS with hardware timestamp

2.7 Interfaces: USB

- One USB 2.0 high-speed interface that can be configured to work in host or device mode
- System clock which serves as the reference clock to derive the PHY clock
- Backward compatible to USB 1.1 full speed
- Power-on configuration bit can be used during power-up to configure the USB interface for host / device operation

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- Power-on boot support in USB device mode (using external USB host)

2.8 Interfaces: SD / SDIO

- Smart media input / output (SMIO) pins for SD controllers
- Power-on eMMC boot modes
- Two SD controllers for SD, SDIO, SDHC, SDXC, MMC, and eMMC memory cards
  - 32 GB maximum capacity for SDHC
  - 1 TB maximum for SDXC
  - The controllers support SD, SDIO, SDHC, SDXC, MMC, and eMMC functions
  - SD0 supports eMMC boot mode
  - SD0 supports 1-, 4-, and 8-bit data bus width
  - SDIO0 supports 1- and 4-bit data bus width

Controller	Capacity			UHS-1	MMC / eMMC	Boot Support	SDIO (1.0 or 3.0)
	SD	SDHC	SDXC				
SD0	√	√	√	√	√	√	3.0
SDIO0	√	√	√	√	√		3.0

Table 2-1. Supported SD Modes; Where √ = Supported.

2.9 Interfaces: SSI / SPI

The CV22FS55 is comprised of nine SSI / SPI interfaces which include six SSI masters, one SSI slave, one SPI NOR, and one SPI NAND.

2.9.1 SSI / SPI Masters

- SSI0 master controls up to eight slave devices
- SSI1 master controls up to four slave devices
- SSI2~SSI5 masters support one slave device each
- Each TX / RX FIFO has 32 entries
- Default device enabled polarity can be inverted with register programming

2.9.2 SSI / SPI Slave

- Each TX / RX FIFO has 32 entries
- Enabled polarity is active low

2.9.3 SPI NOR

- One dedicated SPI controller supports SPI NOR boot

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- The controller supports single- / dual- / quad- / octal-SPI NOR

2.9.4 SPI NAND

- One dedicated SPI controller supports SPI NAND boot
- The controller supports single- / dual- / quad-SPI NAND
- Smart media input / output (SMIO) pins for NAND flash
- Power-on NAND flash boot modes
- NAND flash interface
- Memory systems up to 8 GB and 128 Gb flash chips

2.10 Interfaces: I2C / IDC

- Two-wire, bi-directional bus for data communication between the chip and peripheral devices
- Protocol speeds up to 400 Khz
- Six I2C / IDC masters and one slave interface
- Single master mode

2.11 Interfaces: UART

- Five UART ports
  - 4 AHB-bus UART interfaces, 1 APB-bus UART interface
- Direct memory access (DMA) and hardware flow control in AHB-based UART ports
- Maximum baud rate of 115200 bps (based on per-port software settings)

2.12 Interfaces: General Purpose Input / Output (GPIO)

- 103 CMOS pins which can be programmed for multi-purpose general purpose input / output (GPIO) functions
- Multiplexing with different functions that can be enabled via software

2.13 Interfaces: CAN FD

The CV22FS55 comprises of six CAN FD controllers.

2.14 Interfaces: Real Time Clock (RTC)

- One 32-bit embedded RTC on the CV22FS55 chip:
  - Current time set
  - One dedicated always-on power supply pin (remains active even when the core powers off)

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2.15 Interfaces: Pulse Width Modulator (PWM)

- Three sets of pulse width modulators (PWMs)
  - Each controller supports up to four PWMs
  - Motor control and / or driving LCD panels for video output (various functions enabled via software)

2.16 Interfaces: JTAG

- JTAG-ICE interface support

Contact an Ambarella representative for details on JTAG.

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3. PINS

3.1 Overview of the CV22FS55 Pins

This section provides a list of the 441 external pins according to their location on the CV22FS55 chip. The figure below indicates the orientation of the pins by column (numbers) and row (letters).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
A	VSSI	VSSI	DDR0_DQS_BAR_0_LP4	DDR0_DQS_0_LP4	DDR0_DQ_7_LP4	MIPI_DSI0_DP_3	MIPI_DSI0_DN_0	MIPI_DSI0_DP_1	XIN	ENET1_RXDV	ENET1_CLK_RX	ENET_EXT_OSC_CLK	ENET_GTX_CLK	SD_LVDS_N_6	SD_LVDS_N_5	SD_LVDS_N_4	SD_LVDS_P_3	SPCLK_LVDS_P_0	SD_LVDS_N_0	VSSI	VSSI	A
B	VSSI	DDR0_DQ_9_LP4	DDR0_DQ_8_LP4	DDR0_DQ_0_LP4	DDR0_DQ_6_LP4	MIPI_DSI0_DN_3	MIPI_DSI0_DP_0	MIPI_DSI0_DN_1	XOUT	CEHU_SAFETY_ERR_L_0	ENET_CLK_RX	ENET1_GTX_CLK	ENET_2ND_REF_CLK	SD_LVDS_P_6	SD_LVDS_P_5	SD_LVDS_P_4	SD_LVDS_N_3	SPCLK_LVDS_N_0	SD_LVDS_P_0	DDRE_DQ_1_LP4	VSSI	B
C	DDR0_DM_1_LP4	DDR0_DQ_15_LP4	DDR0_DQ_1_LP4	DDR0_DM_0_LP4	MIPI_DSI1_DP_3	MIPI_DSI1_DP_0	MIPI_DSI1_DP_CLK	MIPI_DSI0_DP_2	MIPI_DSI0_DN_CLK	DETECT_VBUS	ENET1_RXD_0	ENET1_MDC	ENET1_MDIO	ENET1_RXD_3	SD_LVDS_N_7	SPCLK_LVDS_N_1	SD_LVDS_P_2	SD_LVDS_N_1	SD_LVDS_N_11	DDRE_DQ_0_LP4	DDRE_DQS_BAR_0_LP4	C
D	DDR0_DQS_1_LP4	DDR0_DQ_14_LP4	DDR0_DQ_3_LP4	DDR0_DQ_2_LP4	MIPI_DSI1_DN_3	MIPI_DSI1_DN_0	MIPI_DSI1_DN_CLK	MIPI_DSI0_DN_2	MIPI_DSI0_DP_CLK	CAN0_RX	VSSI	ENET1_PTP_PPS	ENET1_TXD_3	ENET1_TXD_1	SD_LVDS_P_7	SPCLK_LVDS_P_1	SD_LVDS_N_2	SD_LVDS_P_1	SD_LVDS_P_11	DDRE_DM_0_LP4	DDRE_DQS_0_LP4	D
E	DDR0_DQS_BAR_1_LP4	DDR0_DQ_13_LP4	DDR0_DQ_5_LP4	DDR0_DQ_4_LP4	MIPI_DSI1_DN_1	JTAG_RST_L	JTAG_TMS	CAN1_RX	CAN0_TX	CAN1_TX	VSSI	ENET1_RXD_2	ENET1_RXD_1	ENET1_TXD_0	ENET1_TXEN	ENET1_TXD_2	ENET_TXD_3	SD_LVDS_P_9	SD_LVDS_N_9	DDRE_DQ_6_LP4	DDRE_DQ_7_LP4	E
F	DDR0_DQ_12_LP4	DDR0_DQ_10_LP4	DDR0_DQ_11_LP4	MIPI_DSI1_DN_2	MIPI_DSI1_DP_1	BG_REXT	OTP_VPP	AVDD	VSSI	VDDP	VSSI	VDDO	VSSI	AVDD18_1	DDRE_DVDD_IO	VSSI	ENET_TXD_2	SPCLK_LVDS_P_2	SPCLK_LVDS_N_2	DDRE_DQ_2_LP4	DDRE_DQ_5_LP4	F
G	DDR0_CA_4_A_LP4	DDR0_CA_1_A_LP4	DDR0_CA_2_A_LP4	MIPI_DSI1_DP_2	JTAG_TDI	VSSI	AVDD18_2	AVDD	VSSI	VDDP	VSSI	VDDO	VSSI	AVDD18_1	DDRE_DVDD_IO	VSSI	ENET_TXD_0	SD_LVDS_P_8	SD_LVDS_N_8	DDRE_DQ_4_LP4	DDRE_DQ_14_LP4	G
H	DDR0_CS_2_A_LP4	DDR0_CA_5_A_LP4	DDR0_CA_3_A_LP4	DDR0_CA_0_A_LP4	JTAG_TDO	VSSI	AVDD18_2	RTC_VDD18	VSSI	VDDI	VSSI	AVDD	VSSI	MIPI_AVDD18_IO	DDRE_DVDD_IO	VSSI	ENET_RXD_1	ENET_TXD_1	SD_LVDS_P_10	DDRE_DQ_3_LP4	DDRE_DM_1_LP4	H
J	DDR0_CK_A_LP4	DDR0_CA_1_B_LP4	DDR0_CS_A_LP4	XO_RTC	JTAG_CLK	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_IO	VSSI	ENET_RXD_3	ENET_MDIO	SD_LVDS_N_10	DDRE_DQ_15_LP4	DDRE_DQ_9_LP4	J
K	DDR0_CK_A_BAR_LP4	DDR0_CA_3_B_LP4	DDR0_CA_0_B_LP4	XI_RTC	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_VDDQ	VSSI	ENET_RXD_2	ENET_RXDV	ENET_TXEN	DDRE_DQ_8_LP4	DDRE_DQS_1_LP4	K
L	DDR0_CK_B_BAR_LP4	DDR0_CA_2_B_LP4	VSSI	VSSI	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_VDDQ	VSSI	ENET_PTP_PPS	ENET_MDC	ENET_RXD_0	DDRE_DQ_10_LP4	DDRE_DQS_BAR_1_LP4	L
M	DDR0_CK_B_LP4	DDR0_CA_4_B_LP4	DDR0_CA_5_B_LP4	VSSI	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_VDDQ	VSSI	VSSI	VSSI	VSSI	DDRE_DQ_13_LP4	DDRE_DQ_11_LP4	M
N	DDR0_CALIBR_LP4	DDR0_CS_2_B_LP4	DDR0_CS_B_LP4	VSSI	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_VDDQ	VSSI	SSI2CLK	SSI3EN0	SSI3MISO	DDRE_DQ_12_LP4	DDRE_CK_A_BAR_LP4	N
P	DDR0_RESET_LP4_LP3	DDR0_CKE_2_LP4_LP3	DDR0_CKE_LP4_LP3	VSSI	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	VDDI	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	DDRE_DVDD_VDDQ	VSSI	CEHU_SAFETY_ERR_L_1	SSI3CLK	SSI3MOSI	DDRE_CA_4_A_LP4	DDRE_CK_A_LP4	P
R	DDR0_DQ_23_LP4	DDR0_DQ_17_LP4	DDR0_DQ_27_LP4	VSSI	DDR_DVDD_VDDQ	VSSI	DDR_DVDD_IO	NAND_VDDIO	VSSI	VDDI	VSSI	VDDI	VSSI	VDDI	VDDO	VSSI	I2S_SI_0	SSI2EN0	GPIO25	DDRE_CA_5_A_LP4	DDRE_CA_3_A_LP4	R
T	DDR0_DQ_21_LP4	DDR0_DQ_20_LP4	DDR0_DQ_28_LP4	VSSI	DDR_DVDD_VDDQ	VSSI	AVDD33	SD_VDDO	VSSI	VDDP	VSSI	AVDD	VSSI	VDDI	VDDO	VSSI	TEST_MODE	SSI2MOSI	SSI2MISO	DDRE_CA_1_A_LP4	DDRE_CA_2_A_LP4	T
U	DDR0_DQ_19_LP4	DDR0_DQ_18_LP4	DDR0_DQ_26_LP4	SMIO_9	SMIO_34	VSSI	VSSI	VSSI	VSSI	VDDP	VSSI	SDIO_VDDO	UART0_AHB_RTS_N	UART_APB_TX	SD_HS_SEL	SSI0CLK	SSI0MISO	SSI0MOSI	SSI0EN1	DDRE_CA_0_A_LP4	DDRE_CS_A_LP4	U
V	DDR0_DQS_BAR_2_LP4	DDR0_DQ_22_LP4	DDR0_DQ_29_LP4	SMIO_8	SMIO_7	VSSI	VSSI	VSSI	VSSI	SENSOR_RST	VSSI	UART1_AHB_TX	UART1_AHB_RX	PWM2	SVSYNC	IDCSCLK	IDCSDATA	I2S_WS	SSI0EN0	DDRE_CKE_LP4_LP3	DDRE_RESET_LP4_LP3	V
W	DDR0_DQS_2_LP4	DDR0_DQ_16_LP4	DDR0_DQ_30_LP4	SMIO_6	SMIO_1	SMIO_0	SMIO_5	SMIO_18	SMIO_23	USB_REXT	VSSI	TIMER0	UART_APB_RX	UART1_AHB_RTS_N	SHSYNC	PWM0	GPIO24	I2S_SO_0	I2S_CLK	CLK_AU	DDRE_CS_2_A_LP4	W
Y	VSSI	DDR0_DM_2_LP4	DDR0_DQ_31_LP4	DDR0_DM_3_LP4	DDR0_DQ_24_LP4	SMIO_35	SMIO_3	SMIO_19	SMIO_22	USB_DP	SMIO_25	SMIO_28	SMIO_26	SMIO_32	SMIO_33	UART0_AHB_TX	UART1_AHB_CTS_N	UART0_AHB_CTS_N	PWM1	POR_L	VSSI	Y
AA	VSSI	VSSI	DDR0_DQS_3_LP4	DDR0_DQS_BAR_3_LP4	DDR0_DQ_25_LP4	SMIO_2	SMIO_4	SMIO_20	SMIO_21	USB_DM	SMIO_24	SMIO_29	SMIO_27	SMIO_31	SMIO_30	CLK_SI_1	CLK_SI	UART0_AHB_RX	DDRE_CALIBR_LP4	VSSI	VSSI	AA
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	

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Figure 3-1. Pin Map for the CV22FS55 Chip.

The following table lists all of the external pins on the CV22FS55 chip in order by map location. Each entry provides the pin name as it appears on the ball map, the location of the pin on the map and on schematics, the functional class, and multiplexed functionality detail if applicable.

Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
A1	VSSI	Ground						
A2	VSSI	Ground						

Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
A3	**DDR0_DQS_BAR_0_LP4**	DDR						
A4	**DDR0_DQS_0_LP4**	DDR						
A5	**DDR0_DQ_7_LP4**	DDR						
A6	**MIPI_DSI0_DP_3**	VOUT						
A7	**MIPI_DSI0_DN_0**	VOUT						
A8	**MIPI_DSI0_DP_1**	VOUT						
A9	**XIN**	Global						
A10	**ENET1_RXDV**	Ethernet	53	ENET1_RXDV	ENET1_RXDV	EXT_VSYNC_SRC_INPUT	SSI4_SCLK	CAN3_RX
A11	**ENET1_CLK_RX**	Ethernet	57	ENET1_REF_CLK	ENET1_CLK_RX		EHCI_APP_PRT_OVCURR0	CAN5_RX
A12	**ENET_EXT_OSC_CLK**	Ethernet	42	ENET_EXT_OSC_CLK	ENET_EXT_OSC_CLK			
A13	**ENET_GTX_CLK**	Ethernet	41	ENET_2ND_REF_CLK	ENET_GTX_CLK	PWM9		
A14	**SD_LVDS_N_6**	VIN						
A15	**SD_LVDS_N_5**	VIN						
A16	**SD_LVDS_N_4**	VIN						
A17	**SD_LVDS_P_3**	VIN						
A18	**SPCLK_LVDS_P_0**	VIN						
A19	**SD_LVDS_N_0**	VIN						
A20	**VSSI**	Ground						
A21	**VSSI**	Ground						
B1	**VSSI**	Ground						
B2	**DDR0_DQ_9_LP4**	DDR						
B3	**DDR0_DQ_8_LP4**	DDR						
B4	**DDR0_DQ_0_LP4**	DDR						
B5	**DDR0_DQ_6_LP4**	DDR						
B6	**MIPI_DSI0_DN_3**	VOUT						
B7	**MIPI_DSI0_DP_0**	VOUT						
B8	**MIPI_DSI0_DN_1**	VOUT						
B9	**XOUT**	Global						
B10	**CEHU_SAFETY_ERR_L_0**	Global						
B11	**ENET_CLK_RX**	Ethernet	40	ENET_REF_CLK	ENET_CLK_RX	PWM8		
B12	**ENET1_GTX_CLK**	Ethernet	58	ENET_2ND_REF_CLK	ENET1_GTX_CLK		EHCI_PRT_PWR_0	CAN5_TX

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
B13	**ENET_2ND_ REF_CLK**	Ethernet	43	ENET_2ND _REF_CLK	ENET_2ND _REF_CLK	WDT_EXT _RST_L	SSI0_EN2	
B14	**SD_LVDS_P_6**	VIN						
B15	**SD_LVDS_P_5**	VIN						
B16	**SD_LVDS_P_4**	VIN						
B17	**SD_LVDS_N_3**	VIN						
B18	**SPCLK_ LVDS_N_0**	VIN						
B19	**SD_LVDS_P_0**	VIN						
B20	**DDRE_DQ_1_ LP4**	DDR						
B21	**VSSI**	Ground						
C1	**DDR0_DM_1_ LP4**	DDR						
C2	**DDR0_DQ_15_ LP4**	DDR						
C3	**DDR0_DQ_1_ LP4**	DDR						
C4	**DDR0_DM_0_ LP4**	DDR						
C5	**MIPI_DSI1_DP_3**	VOUT						
C6	**MIPI_DSI1_DP_0**	VOUT						
C7	**MIPI_DSI1_DP_ CLK**	VOUT						
C8	**MIPI_DSI0_DP_2**	VOUT						
C9	**MIPI_DSI0_DN_ CLK**	VOUT						
C10	**DETECT_VBUS**	USB						
C11	**ENET1_RXD_0**	Ethernet	49	ENET1_ RXD_0	ENET1_ RXD_0	SSI5_SCLK	UART2_ AHB_RX	SSIS_SCLK
C12	**ENET1_MDC**	Ethernet	54	ENET1_ MDC	ENET1_ MDC	AHB_MDC1	SSI4_TXD	CAN3_TX
C13	**ENET1_MDIO**	Ethernet	55	ENET1_ MDIO	ENET1_ MDIO	AHB_MDIO1	SSI4_RXD	CAN4_RX
C14	**ENET1_RXD_3**	Ethernet	52	IDC5DATA	ENET1_ RXD_3	VSDELAY_ VSYNC_ OUT_1	UART2_AHB _RTS_N	SSIS_EN
C15	**SD_LVDS_N_7**	VIN						
C16	**SPCLK_ LVDS_N_1**	VIN						
C17	**SD_LVDS_P_2**	VIN						
C18	**SD_LVDS_N_1**	VIN						
C19	**SD_LVDS_N_11**	VIN						
C20	**DDRE_DQ_0_ LP4**	DDR						
C21	**DDRE_DQS_ BAR_0_LP4**	DDR						
D1	**DDR0_DQS_1_ LP4**	DDR						
D2	**DDR0_DQ_14_ LP4**	DDR						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
D3	**DDR0_DQ_3_LP4**	DDR						
D4	**DDR0_DQ_2_LP4**	DDR						
D5	**MIPI_DSI1_DN_3**	VOUT						
D6	**MIPI_DSI1_DN_0**	VOUT						
D7	**MIPI_DSI1_DN_CLK**	VOUT						
D8	**MIPI_DSI0_DN_2**	VOUT						
D9	**MIPI_DSI0_DP_CLK**	VOUT						
D10	**CAN0_RX**	CAN	98	CAN0_RX				
D11	**VSSI**	Ground						
D12	**ENET1_PTP_PPS**	Ethernet	56	ENET1_PTP_PPS_O	ENET1_PTP_PPS_O	EXT_VSYNC_SRC_INPUT	SSI4_EN0	CAN4_TX
D13	**ENET1_TXD_3**	Ethernet	48		ENET1_TXD_3	VSDELAY_HSYNC_OUT_1		CAN2_TX
D14	**ENET1_TXD_1**	Ethernet	46	ENET1_TXD_1	ENET1_TXD_1	SSI5_EN0		
D15	**SD_LVDS_P_7**	VIN						
D16	**SPCLK_LVDS_P_1**	VIN						
D17	**SD_LVDS_N_2**	VIN						
D18	**SD_LVDS_P_1**	VIN						
D19	**SD_LVDS_P_11**	VIN						
D20	**DDRE_DM_0_LP4**	DDR						
D21	**DDRE_DQS_0_LP4**	DDR						
E1	**DDR0_DQS_BAR_1_LP4**	DDR						
E2	**DDR0_DQ_13_LP4**	DDR						
E3	**DDR0_DQ_5_LP4**	DDR						
E4	**DDR0_DQ_4_LP4**	DDR						
E5	**MIPI_DSI1_DN_1**	VOUT						
E6	**JTAG_RST_L**	JTAG						
E7	**JTAG_TMS**	JTAG						
E8	**CAN1_RX**	CAN	100	CAN1_RX				
E9	**CAN0_TX**	CAN	99	CAN0_TX				
E10	**CAN1_TX**	CAN	101	CAN1_TX				
E11	**VSSI**	Ground						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
E12	**ENET1_RXD_2**	Ethernet	51	IDC5CLK	ENET1_ RXD_2	VSDELAY_ VSYNC_ OUT_0	UART2_ AHB_CTS_N	SSIS_RXD
E13	**ENET1_RXD_1**	Ethernet	50	ENET1_ RXD_1	ENET1_ RXD_1	SSI5_RXD	UART2_ AHB_TX	SSIS_TXD
E14	**ENET1_TXD_0**	Ethernet	45	ENET1_ TXD_0	ENET1_ TXD_0	SSI5_TXD		PWM11
E15	**ENET1_TXEN**	Ethernet	44	ENET1_ TXEN	ENET1_ TXEN			CAN2_RX
E16	**ENET1_TXD_2**	Ethernet	47		ENET1_ TXD_2	VSDELAY_ HSYNC_ OUT_0		
E17	**ENET_TXD_3**	Ethernet	31		ENET_ TXD_3			
E18	**SD_LVDS_P_9**	VIN						
E19	**SD_LVDS_N_9**	VIN						
E20	**DDRE_DQ_6_ LP4**	DDR						
E21	**DDRE_DQ_7_ LP4**	DDR						
F1	**DDR0_DQ_12_ LP4**	DDR						
F2	**DDR0_DQ_10_ LP4**	DDR						
F3	**DDR0_DQ_11_ LP4**	DDR						
F4	**MIPI_DSI1_ DN_2**	VOUT						
F5	**MIPI_DSI1_DP_1**	VOUT						
F6	**BG_REXT**	Global						
F7	**OTP_VPP**	OTP						
F8	**AVDD**	Power						
F9	**VSSI**	Ground						
F10	**VDDP**	Power						
F11	**VSSI**	Ground						
F12	**VDDO**	Power						
F13	**VSSI**	Ground						
F14	**AVDD18_1**	Power						
F15	**DDRE_DVDD_IO**	DDR						
F16	**VSSI**	Ground						
F17	**ENET_TXD_2**	Ethernet	30		ENET_ TXD_2	PWM11		
F18	**SPCLK_ LVDS_P_2**	VIN						
F19	**SPCLK_ LVDS_N_2**	VIN						
F20	**DDRE_DQ_2_ LP4**	DDR						
F21	**DDRE_DQ_5_ LP4**	DDR						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
G1	**DDR0_CA_4_A_LP4**	DDR						
G2	**DDR0_CA_1_A_LP4**	DDR						
G3	**DDR0_CA_2_A_LP4**	DDR						
G4	**MIPI_DSI1_DP_2**	VOUT						
G5	**JTAG_TDI**	JTAG						
G6	**VSSI**	Ground						
G7	**AVDD18_2**	Power						
G8	**AVDD**	Power						
G9	**VSSI**	Ground						
G10	**VDDP**	Power						
G11	**VSSI**	Ground						
G12	**VDDO**	Power						
G13	**VSSI**	Ground						
G14	**AVDD18_1**	Power						
G15	**DDRE_DVDD_IO**	DDR						
G16	**VSSI**	Ground						
G17	**ENET_TXD_0**	Ethernet	28	ENET_TXD_0	ENET_TXD_0	PWM9		
G18	**SD_LVDS_P_8**	VIN						
G19	**SD_LVDS_N_8**	VIN						
G20	**DDRE_DQ_4_LP4**	DDR						
G21	**DDRE_DQ_14_LP4**	DDR						
H1	**DDR0_CS_2_A_LP4**	DDR						
H2	**DDR0_CA_5_A_LP4**	DDR						
H3	**DDR0_CA_3_A_LP4**	DDR						
H4	**DDR0_CA_0_A_LP4**	DDR						
H5	**JTAG_TDO**	JTAG						
H6	**VSSI**	Ground						
H7	**AVDD18_2**	Power						
H8	**RTC_VDD18**	Power						
H9	**VSSI**	Ground						
H10	**VDDI**	Power						
H11	**VSSI**	Ground						
H12	**AVDD**	Power						
H13	**VSSI**	Ground						
H14	**MIPI_AVDD18_IO**	Power						
H15	**DDRE_DVDD_IO**	DDR						
H16	**VSSI**	Ground						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
H17	**ENET_RXD_1**	Ethernet	33	ENET_ RXD_1	ENET_ RXD_1			
H18	**ENET_TXD_1**	Ethernet	29	ENET_ TXD_1	ENET_ TXD_1	PWM10		
H19	**SD_LVDS_P_10**	VIN						
H20	**DDRE_DQ_3_ LP4**	DDR						
H21	**DDRE_DM_1_ LP4**	DDR						
J1	**DDR0_CK_A_ LP4**	DDR						
J2	**DDR0_CA_1_B_ LP4**	DDR						
J3	**DDR0_CS_A_ LP4**	DDR						
J4	**XO_RTC**	RTC						
J5	**JTAG_CLK**	JTAG						
J6	**VSSI**	Ground						
J7	**DDR_DVDD_IO**	DDR						
J8	**VDDI**	Power						
J9	**VSSI**	Ground						
J10	**VDDI**	Power						
J11	**VSSI**	Ground						
J12	**VDDI**	Power						
J13	**VSSI**	Ground						
J14	**VDDI**	Power						
J15	**DDRE_DVDD_IO**	DDR						
J16	**VSSI**	Ground						
J17	**ENET_RXD_3**	Ethernet	35		ENET_ RXD_3	PWM3		
J18	**ENET_MDIO**	Ethernet	38	ENET_MDIO	ENET_MDIO	PWM6	AHB_MDIO	
J19	**SD_LVDS_N_10**	VIN						
J20	**DDRE_DQ_15_ LP4**	DDR						
J21	**DDRE_DQ_9_ LP4**	DDR						
K1	**DDR0_CK_A_ BAR_LP4**	DDR						
K2	**DDR0_CA_3_B_ LP4**	DDR						
K3	**DDR0_CA_0_B_ LP4**	DDR						
K4	**XI_RTC**	RTC						
K5	**DDR_DVDD_ VDDQ**	Power						
K6	**VSSI**	Ground						
K7	**DDR_DVDD_IO**	DDR						
K8	**VDDI**	Power						
K9	**VSSI**	Ground						
K10	**VDDI**	Power						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
K11	**VSSI**	Ground						
K12	**VDDI**	Power						
K13	**VSSI**	Ground						
K14	**VDDI**	Power						
K15	**DDRE_DVDD_VDDQ**	Power						
K16	**VSSI**	Ground						
K17	**ENET_RXD_2**	Ethernet	34		ENET_RXD_2			
K18	**ENET_RXDV**	Ethernet	36	ENET_RXDV	ENET_RXDV	PWM4		
K19	**ENET_TXEN**	Ethernet	27	ENET_TXEN	ENET_TXEN	PWM8		
K20	**DDRE_DQ_8_LP4**	DDR						
K21	**DDRE_DQS_1_LP4**	DDR						
L1	**DDR0_CK_B_BAR_LP4**	DDR						
L2	**DDR0_CA_2_B_LP4**	DDR						
L3	**VSSI**	Ground						
L4	**VSSI**	Ground						
L5	**DDR_DVDD_VDDQ**	Power						
L6	**VSSI**	Ground						
L7	**DDR_DVDD_IO**	DDR						
L8	**VDDI**	Power						
L9	**VSSI**	Ground						
L10	**VDDI**	Power						
L11	**VSSI**	Ground						
L12	**VDDI**	Power						
L13	**VSSI**	Ground						
L14	**VDDI**	Power						
L15	**DDRE_DVDD_VDDQ**	Power						
L16	**VSSI**	Ground						
L17	**ENET_PTP_PPS**	Ethernet	39	ENET_PTP_PPS_O	ENET_PTP_PPS_O	PWM7		
L18	**ENET_MDC**	Ethernet	37	ENET_MDC	ENET_MDC	PWM5	AHB_MDC	
L19	**ENET_RXD_0**	Ethernet	32	ENET_RXD_0	ENET_RXD_0			
L20	**DDRE_DQ_10_LP4**	DDR						
L21	**DDRE_DQS_BAR_1_LP4**	DDR						
M1	**DDR0_CK_B_LP4**	DDR						
M2	**DDR0_CA_4_B_LP4**	DDR						
M3	**DDR0_CA_5_B_LP4**	DDR						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
M4	**VSSI**	Ground						
M5	**DDR_DVDD_ VDDQ**	Power						
M6	**VSSI**	Ground						
M7	**DDR_DVDD_IO**	DDR						
M8	**VDDI**	Power						
M9	**VSSI**	Ground						
M10	**VDDI**	Power						
M11	**VSSI**	Ground						
M12	**VDDI**	Power						
M13	**VSSI**	Ground						
M14	**VDDI**	Power						
M15	**DDRE_DVDD_ VDDQ**	Power						
M16	**VSSI**	Ground						
M17	**VSSI**	Ground						
M18	**VSSI**	Ground						
M19	**VSSI**	Ground						
M20	**DDRE_DQ_13_ LP4**	DDR						
M21	**DDRE_DQ_11_ LP4**	DDR						
N1	**DDR0_CALIBR_ LP4**	DDR						
N2	**DDR0_CS_2_B_ LP4**	DDR						
N3	**DDR0_CS_B_ LP4**	DDR						
N4	**VSSI**	Ground						
N5	**DDR_DVDD_ VDDQ**	Power						
N6	**VSSI**	Ground						
N7	**DDR_DVDD_IO**	DDR						
N8	**VDDI**	Power						
N9	**VSSI**	Ground						
N10	**VDDI**	Power						
N11	**VSSI**	Ground						
N12	**VDDI**	Power						
N13	**VSSI**	Ground						
N14	**VDDI**	Power						
N15	**DDRE_DVDD_ VDDQ**	Power						
N16	**VSSI**	Ground						
N17	**SSI2CLK**	SSI	8	SSI2_SCLK	IDC2CLK	SSI0_EN2	VSDELAY_ VSYNC_ OUT_0	
N18	**SSI3EN0**	SSI	15	SSI3_EN0	IDC5DATA	SSIS_EN	UART3_ AHB_RTS_N	
N19	**SSI3MISO**	SSI	14	SSI3_RXD	IDC5CLK	SSIS_RXD	UART3_ AHB_CTS_N	

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
N20	**DDRE_DQ_12_LP4**	DDR						
N21	**DDRE_CK_A_BAR_LP4**	DDR						
P1	**DDR0_RESET_LP4_LP3**	DDR						
P2	**DDR0_CKE_2_LP4_LP3**	DDR						
P3	**DDR0_CKE_LP4_LP3**	DDR						
P4	**VSSI**	Ground						
P5	**DDR_DVDD_VDDQ**	Power						
P6	**VSSI**	Ground						
P7	**DDR_DVDD_IO**	DDR						
P8	**VDDI**	Power						
P9	**VSSI**	Ground						
P10	**VDDI**	Power						
P11	**VSSI**	Ground						
P12	**VDDI**	Power						
P13	**VSSI**	Ground						
P14	**VDDI**	Power						
P15	**DDRE_DVDD_VDDQ**	Power						
P16	**VSSI**	Ground						
P17	**CEHU_SAFETY_ERR_L_1**	Global						
P18	**SSI3CLK**	SSI	12	SSI3_SCLK	IDC4CLK	SSIS_SCLK	UART3_AHB_RX	
P19	**SSI3MOSI**	SSI	13	SSI3_TXD	IDC4DATA	SSIS_TXD	UART3_AHB_TX	
P20	**DDRE_CA_4_A_LP4**	DDR						
P21	**DDRE_CK_A_LP4**	DDR						
R1	**DDR0_DQ_23_LP4**	DDR						
R2	**DDR0_DQ_17_LP4**	DDR						
R3	**DDR0_DQ_27_LP4**	DDR						
R4	**VSSI**	Ground						
R5	**DDR_DVDD_VDDQ**	Power						
R6	**VSSI**	Ground						
R7	**DDR_DVDD_IO**	DDR						
R8	**NAND_VDDO**	Power						
R9	**VSSI**	Ground						
R10	**VDDI**	Power						
R11	**VSSI**	Ground						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
R12	**VDDI**	Power						
R13	**VSSI**	Ground						
R14	**VDDI**	Power						
R15	**VDDO**	Power						
R16	**VSSI**	Ground						
R17	**I2S_SI_0**	I2S	22	I2S_SI_0	SSI1_TXD			
R18	**SSI2EN0**	SSI	11	SSI2_EN0	IDC3DATA	SSI0_EN5	VSDELAY_HSYNC_OUT_1	
R19	**GPIO25**	I2S	25			EHCI_PRT_PWR_0		
R20	**DDRE_CA_5_A_LP4**	DDR						
R21	**DDRE_CA_3_A_LP4**	DDR						
T1	**DDR0_DQ_21_LP4**	DDR						
T2	**DDR0_DQ_20_LP4**	DDR						
T3	**DDR0_DQ_28_LP4**	DDR						
T4	**VSSI**	Ground						
T5	**DDR_DVDD_VDDQ**	Power						
T6	**VSSI**	Ground						
T7	**AVDD33**	Power						
T8	**SD_VDDO**	Power						
T9	**VSSI**	Ground						
T10	**VDDP**	Power						
T11	**VSSI**	Ground						
T12	**AVDD**	Power						
T13	**VSSI**	Ground						
T14	**VDDI**	Power						
T15	**VDDO**	Power						
T16	**VSSI**	Ground						
T17	**TEST_MODE**	Global						
T18	**SSI2MOSI**	SSI	9	SSI2_TXD	IDC2DATA	SSI0_EN3	VSDELAY_VSYNC_OUT_1	
T19	**SSI2MISO**	SSI	10	SSI2_RXD	IDC3CLK	SSI0_EN4	VSDELAY_HSYNC_OUT_0	
T20	**DDRE_CA_1_A_LP4**	DDR						
T21	**DDRE_CA_2_A_LP4**	DDR						
U1	**DDR0_DQ_19_LP4**	DDR						
U2	**DDR0_DQ_18_LP4**	DDR						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
U3	**DDR0_DQ_26_LP4**	DDR						
U4	**SMIO_9**	SMIO	68		SNAND_SI[3]	NORSPI_DQ[3]		
U5	**SMIO_34**	SMIO	85			NORSPI_DQS		
U6	**VSSI**	Ground						
U7	**VSSI**	Ground						
U8	**VSSI**	Ground						
U9	**VSSI**	Ground						
U10	**VDDP**	Power						
U11	**VSSI**	Ground						
U12	**SDIO_VDDO**	Power						
U13	**UART0_AHB_RTS_N**	UART	93	UART0_AHB_RTS_N	SSI1_EN0			TM13_CLK
U14	**UART_APB_TX**	UART	17	UART_APB_TX	EHCI_PRT_PWR_0			
U15	**SD_HS_SEL**	GPIO	102	SD_HS_SEL				
U16	**SSI0CLK**	SSI	3	SSI0_SCLK	IDC0CLK	PWM4		
U17	**SSI0MISO**	SSI	5	SSI0_RXD	IDC1CLK	PWM6		
U18	**SSI0MOSI**	SSI	4	SSI0_TXD	IDC0DATA	PWM5		
U19	**SSI0EN1**	SSI	7	SSI0_EN1		WDT_EXT_RST_L	SDIO0_HS_SEL	
U20	**DDRE_CA_0_A_LP4**	DDR						
U21	**DDRE_CS_A_LP4**	DDR						
V1	**DDR0_DQS_BAR_2_LP4**	DDR						
V2	**DDR0_DQ_22_LP4**	DDR						
V3	**DDR0_DQ_29_LP4**	DDR						
V4	**SMIO_8**	SMIO	67		SNAND_SI[2]	NORSPI_DQ[2]		
V5	**SMIO_7**	SMIO	66		SNAND_SI[1]	NORSPI_DQ[1]		
V6	**VSSI**	Ground						
V7	**VSSI**	Ground						
V8	**VSSI**	Ground						
V9	**VSSI**	Ground						
V10	**SENSOR_RST**	VIN	87		WDT_EXT_RST_L			
V11	**VSSI**	Ground						
V12	**UART1_AHB_TX**	UART	95	UART1_AHB_TX				
V13	**UART1_AHB_RX**	UART	94	UART1_AHB_RX	TM12_CLK			
V14	**PWM2**	PWM	20	PWM2	WDT_EXT_RST_L			

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
V15	**SVSYNC**	VIN	89	IDSP_VIN0_IOPAD_MASTER_VSYNC	IDSP_VIN1_IOPAD_MASTER_VSYNC			
V16	**IDCSCLK**	I2C	1	IDCSCLK	TM12_CLK	SSI0_EN6	VIN_STRIG0	EHCI_APP_PRT_OV-CURR0
V17	**IDCSDATA**	I2C	2	IDCSDATA	TM13_CLK	SSI0_EN7	VIN_STRIG1	EHCI_PRT_PWR_0
V18	**I2S_WS**	I2S	26	I2S_WS				
V19	**SSI0EN0**	SSI	6	SSI0_EN0	IDC1DATA	PWM7		
V20	**DDRE_CKE_LP4_LP3**	DDR						
V21	**DDRE_RESET_LP4_LP3**	DDR						
W1	**DDR0_DQS_2_LP4**	DDR						
W2	**DDR0_DQ_16_LP4**	DDR						
W3	**DDR0_DQ_30_LP4**	DDR						
W4	**SMIO_6**	SMIO	65		SNAND_SI[0]	NORSPI_DQ[0]		
W5	**SMIO_1**	SMIO	60		SNAND_CS_N	NORSPI_EN		
W6	**SMIO_0**	SMIO	59		SNAND_SCK	NORSPI_CLK		
W7	**SMIO_5**	SMIO	64		SD_WP			
W8	**SMIO_18**	SMIO	69		SD_D[0]			
W9	**SMIO_23**	SMIO	74		SD_D[5]	NORSPI_DQ[5]		
W10	**USB_REXT**	USB						
W11	**VSSI**	Ground						
W12	**TIMER0**	Timer	0	TM11_CLK	WDT_EXT_RST_L	PWM3		SDIO0_HS_SEL
W13	**UART_APB_RX**	UART	16	UART_APB_RX	EHCI_APP_PRT_OV-CURR0			
W14	**UART1_AHB_RTS_N**	UART	97	UART1_AHB_RTS_N	IDSP_VIN1_IOPAD_MASTER_HSYNC			
W15	**SHSYNC**	VIN	88	IDSP_VIN0_IOPAD_MASTER_HSYNC	IDSP_VIN1_IOPAD_MASTER_HSYNC			
W16	**PWM0**	PWM	18	PWM0	VIN_STRIG0			
W17	**GPIO24**	I2S	24		SSI1_RXD	EHCI_APP_PRT_OV-CURR0		
W18	**I2S_SO_0**	I2S	23	I2S_SO_0	SSI1_EN0			
W19	**I2S_CLK**	I2S	21	I2S_CLK	SSI1_SCLK			
W20	**CLK_AU**	I2S						
W21	**DDRE_CS_2_A_LP4**	DDR						

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
Y1	**VSSI**	Ground						
Y2	**DDR0_DM_2_LP4**	DDR						
Y3	**DDR0_DQ_31_LP4**	DDR						
Y4	**DDR0_DM_3_LP4**	DDR						
Y5	**DDR0_DQ_24_LP4**	DDR						
Y6	**SMIO_35**	SMIO	86		SD_RESET			
Y7	**SMIO_3**	SMIO	62		SD_CMD			
Y8	**SMIO_19**	SMIO	70		SD_D[1]			
Y9	**SMIO_22**	SMIO	73		SD_D[4]	NORSPI_DQ[4]		
Y10	**USB_DP**	USB						
Y11	**SMIO_25**	SMIO	76		SD_D[7]	NORSPI_DQ[7]		
Y12	**SMIO_28**	SMIO	79	UART2_AHB_CTS_N	SDIO0_D[0]	SSI1_RXD	SNAND_SI[0]	NORSPI_DQ[0]
Y13	**SMIO_26**	SMIO	77	UART2_AHB_RX	SDIO0_CLK	SSI1_SCLK	SNAND_SCK	NORSPI_CLK
Y14	**SMIO_32**	SMIO	83	UART3_AHB_RX	SDIO0_CD	SSI1_EN3		
Y15	**SMIO_33**	SMIO	84	UART3_AHB_TX	SDIO0_WP			
Y16	**UART0_AHB_TX**	UART	91	UART0_AHB_TX	SSI1_TXD			WDT_EXT_RST_L
Y17	**UART1_AHB_CTS_N**	UART	96	UART1_AHB_CTS_N	IDSP_VIN1_IOPAD_MASTER_VSYNC			
Y18	**UART0_AHB_CTS_N**	UART	92	UART0_AHB_CTS_N	SSI1_RXD	EHCI_PRT_PWR_0		TM12_CLK
Y19	**PWM1**	PWM	19	PWM1	VIN_STRIG1			
Y20	**POR_L**	Global						
Y21	**VSSI**	Ground						
AA1	**VSSI**	Ground						
AA2	**VSSI**	Ground						
AA3	**DDR0_DQS_3_LP4**	DDR						
AA4	**DDR0_DQS_BAR_3_LP4**	DDR						
AA5	**DDR0_DQ_25_LP4**	DDR						
AA6	**SMIO_2**	SMIO	61		SD_CLK			
AA7	**SMIO_4**	SMIO	63		SD_CD			
AA8	**SMIO_20**	SMIO	71		SD_D[2]			
AA9	**SMIO_21**	SMIO	72		SD_D[3]			
AA10	**USB_DM**	USB						
AA11	**SMIO_24**	SMIO	75		SD_D[6]	NORSPI_DQ[6]		

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Loc.	Pin Name	Class	Multiplexed Functions					
			GPIO	First	Second	Third	Fourth	Fifth
AA12	**SMIO_29**	SMIO	80	UART2_ AHB_RTS_N	SDIO0_D[1]	SSI1_EN0	SNAND_SI[1]	NORSPI_ DQ[1]
AA13	**SMIO_27**	SMIO	78	UART2_ AHB_TX	SDIO0_CMD	SSI1_TXD	SNAND_ CS_N	NORSPI_EN
AA14	**SMIO_31**	SMIO	82	UART3_ AHB_RTS_N	SDIO0_D[3]	SSI1_EN2	SNAND_SI[3]	NORSPI_ DQ[3]
AA15	**SMIO_30**	SMIO	81	UART3_ AHB_CTS_N	SDIO0_D[2]	SSI1_EN1	SNAND_SI[2]	NORSPI_ DQ[2]
AA16	**CLK_SI_1**	VIN						
AA17	**CLK_SI**	VIN						
AA18	**UART0_AHB_RX**	UART	90	UART0_ AHB_RX	SSI1_SCLK	EHCI_APP_ PRT_OV-CURR0	TM11_CLK	PWM10
AA19	**DDRE_CALIBR_ LP4**	DDR						
AA20	**VSSI**	Ground						
AA21	**VSSI**	Ground						

Table 3-1. Pin List and Mapping Table for the CV22FS55 Chip.

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3.2 Pin Tables

The pins for the CV22FS55 chip are classified according to interface as follows:

- (Section 3.2.1) Pins: DRAM for Data
- (Section 3.2.2) Pins: DRAM for ECC
- (Section 3.2.3) Pins: Sensor / Video Input
- (Section 3.2.4) VOUT Pins: MIPI DSI
- (Section 3.2.5) Pins: I2S Digital Audio
- (Section 3.2.6) Pins: Gigabit Ethernet Interface
- (Section 3.2.7) Pins: USB
- (Section 3.2.8) Pins: CAN BUS
- (Section 3.2.9) Pins: Smart Media Input / Output (SMIO)
- (Section 3.2.10) Pins: SSI / SPI
- (Section 3.2.11) Pins: I2C / IDC
- (Section 3.2.12) Pins: UART
- (Section 3.2.13) Pins: General Purpose Input / Output (GPIO)
- (Section 3.2.14) Pins: Real Time Clock (RTC)
- (Section 3.2.15) Pins: Timer
- (Section 3.2.16) Pins: Pulse Width Modulator (PWM)
- (Section 3.2.17) Pins: JTAG Control
- (Section 3.2.18) Pins: Global and Test
- (Section 3.2.19) Pins: Safety Error Status
- (Section 3.2.20) Pins: Power, Ground, and PLL

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3.2.1 Pins: DRAM for Data

This section includes all the 32-bit data bus pins for data.

Name	Location	Dir	Type	Description
DDR0_CA_[0:5]_A_LP4	H4, G2, G3, H3, G1, H2	O	LVSTL	LPDDR4 command/address
DDR0_CA_[0:5]_B_LP4	K3, J2, L2, K2, M2, M3			
DDR0_CALIBR_LP4	N1	I/O	Analog	Pull up to VDDQ via a 240-ohm resistor
DDR0_CK_A_LP4	J1	O	LVSTL	DRAM clock per SDRAM
DDR0_CK_B_LP4	M1			
DDR0_CK_A_BAR_LP4	K1	O	LVSTL	**DDR0_CK_A/B_LP4** and **DDR0_CK_A/B_BAR_LP4** are differential clocks
DDR0_CK_B_BAR_LP4	L1			
DDR0_CS_A_LP4	J3	O	LVSTL	Chip select
DDR0_CS_B_LP4	N3			
DDR0_CS_2_A_LP4	H1			
DDR0_CS_2_B_LP4	N2			

Name	Location	Dir	Type	Description
DDR0_DM_[0:3]_LP4	C4, C1, Y2, Y4	O	LVSTL	Data write mask (1 bit per 8 data bits)
DDR0_DQ_[0:31]_LP4	B4, C3, D4, D3, E4, E3, B5, A5, B3, B2, F2, F3, F1, E2, D2, C2, W2, R2, U2, U1, T2, T1, V2, R1, Y5, AA5, U3, R3, T3, V3,W3, Y3	I/O	LVSTL	Bi-directional data bus
DDR0_DQS_[0:3]_LP4	A4, D1, W1, AA3	I/O	LVSTL	Data strobe (1 bit per 8 data bits) Output with write data, center aligned Input with read data, edge aligned
DDR0_DQS_BAR_[0:3]_LP4	A3, E1, V1, AA4	I/O	LVSTL	**DDR0_DQS_[0:3]_LP4** and **DDR0_DQS_BAR_[0:3]_LP4** are differential signals
DDR0_RESET_LP4_LP3	P1	O	LVSTL	DRAM reset
DDR0_CKE_LP4_LP3	P3	O	LVSTL	DRAM clock enable
DDR0_CKE_1_LP4_LP3	P2	O	LVSTL	DRAM clock enable

Table 3-2. DRAM Pins.

3.2.2 Pins: DRAM for ECC

This section includes all the 16-bit data bus pins for ECC.

Name	Location	Dir	Type	Description
DDRE_CA_[0:5]_A_LP4	U20, T20, T21, R21, P20, R20	O	LVSTL	LPDDR4 command/address
DDRE_CALIBR_LP4	AA19	I/O	Analog	Pull up to VDDQ via a 240-ohm resistor
DDRE_CK_A_LP4	P21	O	LVSTL	DRAM clock per SDRAM
DDRE_CK_A_BAR_LP4	N21	O	LVSTL	**DDR0_CK_A/B_LP4** and **DDR0_CK_A/B_BAR_LP4** are differential clocks
DDRE_CS_A_LP4	U21	O	LVSTL	Chip select
DDRE_CS_2_A_LP4	W21			
DDRE_DM_[0:1]_LP4	D20, H21	O	LVSTL	Data write mask (1 bit per 8 data bits)
DDRE_DQ_[0:15]_LP4	C20, B20, F20, H20, G20, F21, E20, E21, K20, J21, L20, M21, N20, M20, G21, J20,	I/O	LVSTL	Bi-directional data bus
DDRE_DQS_[0:1]_LP4	D21, K21	I/O	LVSTL	Data strobe (1 bit per 8 data bits) Output with write data, center aligned Input with read data, edge aligned
DDRE_DQS_BAR_[0:1]_LP4	C21, L21	I/O	LVSTL	**DDR0_DQS_[0:3]_LP4** and **DDR0_DQS_BAR_[0:3]_LP4** are differential signals
DDRE_RESET_LP4_LP3	V21	O	LVSTL	DRAM reset
DDRE_CKE_LP4_LP3	V20	O	LVSTL	DRAM clock enable

Table 3-3. DRAM Pins.

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3.2.3 Pins: Sensor / Video Input

Name	Location	Dir	Type	Description
CLK_SI	AA17	I/O	CMOS	Sensor clock out
CLK_SI_1	AA16			
SD_LVDS_N_[0:11]	A19, C18, D17, B17, A16, A15, A14, C15, G19, E19, J19, C19	I	MIPI	Sensor data Differential for MIPI
SD_LVDS_P_[0:11]	B19, D18, C17, A17, B16, B15, B14, D15, G18, E18, H19, D19	I	MIPI	
SHSYNC	W15	O	CMOS	H-Sync / H-Valid with Master mode configuration
SPCLK_LVDS_N_[0:2]	B18, C16, F19	I	MIPI	Sensor pixel clock Differential pairs for MIPI
SPCLK_LVDS_P_[0:2]	A18, D16, F18	I	MIPI	
SVSYNC	V15	O	CMOS	V-Sync / V-Valid with Master mode configuration
SENSOR_RST	V10	I/O	CMOS	Sensor reset

Table 3-4. VIN Sensor Interface Pins.

Notes:

- Pins SD_LVDS_N/P_[0:3] support the MIPI virtual channel.
- Refer to the Video Input chapter of the *CV22FS55 System Hardware* document for the MIPI CSI-2 pin map.

3.2.4 VOUT Pins: MIPI DSI

Name	Location	Dir	Type	Description
MIPI_DSI[0:1]_DN_CLK	C9, D7	I/O	Analog	DSI Clock
MIPI_DSI[0:1]_DP_CLK	D9, C7	I/O	Analog	DSI Clock
MIPI_DSI0_DN_[0:3]	A7, B8, D8, B6	I/O	Analog	DSI Data
MIPI_DSI0_DP_[0:3]	B7, A8, C8, A6	I/O	Analog	DSI Data
MIPI_DSI1_DN_[0:3]	D6, E5, F4, D5	I/O	Analog	DSI Data
MIPI_DSI1_DP_[0:3]	C6, F5, G4, C5	I/O	Analog	DSI Data

Table 3-5. MIPI DSI Pins.

3.2.5 Pins: I2S Digital Audio

Name	Location	Dir	Type	Description
CLK_AU	W20	O	CMOS	I2S audio master clock
I2S_CLK	W19	I/O	CMOS	I2S audio bit clock
I2S_SI_0	R17	I	CMOS	I2S serial data in
I2S_SO_0	W18	O	CMOS	I2S serial data out
I2S_WS	V19	I/O	CMOS	I2S word select

Table 3-6. I2S Controller Pins.

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3.2.6 Pins: Gigabit Ethernet Interface

Name	Location	Dir	Type	Description
ENET_CLK_RX	B11	I/O	CMOS	Receive clock
ENET_PTP_PPS	L17	I/O	CMOS	Pulse-per-second output
ENET_EXT_OSC_CLK	A12	I/O	CMOS	External clock source for ENET_GTX_CLK and ENET1_GTX_CLK
ENET_GTX_CLK	A13	I/O	CMOS	Ethernet transmit clock
ENET_MDC	L18	I/O	CMOS	Management data clock
ENET_MDIO	J18	I/O	CMOS	Management data input / output
ENET_RXD_0	L19	I/O	CMOS	Receive data
ENET_RXD_1	H17	I/O	CMOS	
ENET_RXD_2	K17	I/O	CMOS	
ENET_RXD_3	J17	I/O	CMOS	
ENET_RXDV	K18	I/O	CMOS	Receive data valid
ENET_TXD_0	G17	I/O	CMOS	Transmit data
ENET_TXD_1	H18	I/O	CMOS	
ENET_TXD_2	F17	I/O	CMOS	
ENET_TXD_3	E17	I/O	CMOS	
ENET_TXEN	K19	I/O	CMOS	Transmit ready
ENET1_CLK_RX	A11	I/O	CMOS	Receive clock
ENET_2ND_REF_CLK	B13	I/O	CMOS	Reference clock output for external Ethernet PHY
ENET1_PTP_PPS	D12	I/O	CMOS	Pulse-per-second output
ENET1_GTX_CLK	B12	I/O	CMOS	Ethernet transmit clock
ENET1_MDC	C12	I/O	CMOS	Management data clock
ENET1_MDIO	C13	I/O	CMOS	Management data input / output
ENET1_RXD_0	C11	I/O	CMOS	Receive data
ENET1_RXD_1	E13	I/O	CMOS	
ENET1_RXD_2	E12	I/O	CMOS	
ENET1_RXD_3	C14	I/O	CMOS	
ENET1_RXDV	A10	I/O	CMOS	Receive data valid
ENET1_TXD_0	E14	I/O	CMOS	Transmit data
ENET1_TXD_1	D14	I/O	CMOS	
ENET1_TXD_2	E16	I/O	CMOS	
ENET1_TXD_3	D13	I/O	CMOS	
ENET1_TXEN	E15	I/O	CMOS	Transmit ready

Table 3-7. Gigabit Ethernet Pins.

3.2.7 Pins: USB

Name	Location	Dir	Type	Description
DETECT_VBUS	C10	I/O	CMOS	USB slave bus detect
USB_DM	AA10	I/O	Analog	Data
USB_DP	Y10	I/O	Analog	Data
USB_REXT	W10	I/O	Analog	Reference resistor

Table 3-8. USB Interface Pins.

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3.2.8 Pins: CAN BUS

Name	Location	Dir	Type	Description
CAN0_TX	E9	O	CMOS	Controller Area Network 0 - Transmit Data Output
CAN0_RX	D10	I	CMOS	Controller Area Network 0 - Receive Data Input
CAN1_TX	E10	O	CMOS	Controller Area Network 1 - Transmit Data Output
CAN1_RX	E8	I	CMOS	Controller Area Network 1 - Receive Data Input

Table 3-9. CAN Bus Interface Pins.

3.2.9 Pins: Smart Media Input / Output (SMIO)

- The smart media input / output (SMIO) pins are CMOS type and programmable input / output.
- The pins are shared by controllers for boot flash SPI NOR / SPI NAND and SD / SDIO / SDHC / SDXC / MMC / eMMC (SD).
- These pins use **SMIO_[N]** for the primary function name.
- The SMIO interface shares pins with general purpose input / output (GPIO) in Section 3.2.13. There also are GPIO pins that handle smart media functions (i.e., without sharing with SMIO pins). These are listed in the SMIO tables below.

Name	Loc.	SPI NOR		SPI NAND		SD		Description
		Function	Dir	Function	Dir	Function	Dir	
SMIO_0	W6	NORSPI_CLK	O	SNAND_SCK	O			SPI NOR / SPI NAND clock
SMIO_1	W5	NORSPI_EN	O	SNAND_CS_N	O			SPI NOR enable / SPI NAND chip select
SMIO_2	AA6					SD_CLK	O	SD0 clock
SMIO_3	Y7					SD_CMD	I/O	SD0 command
SMIO_4	AA7					SD_CD	I	SD0 card detect
SMIO_5	W7					SD_WP	O	SD0 write protect
SMIO_6	W4	NORSPI_DQ[0]	I/O	SNAND_SI[0]	I/O			SPI NOR / SPI NAND data
SMIO_7	V5	NORSPI_DQ[1]	I/O	SNAND_SI[1]	I/O			SPI NAND / SPI NOR data
SMIO_8	V4	NORSPI_DQ[2]	I/O	SNAND_SI[2]	I/O			SPI NAND / SPI NOR data
SMIO_9	U4	NORSPI_DQ[3]	I/O	SNAND_SI[3]	I/O			SPI NAND / SPI NOR data
SMIO_18	W8					SD_D[0]	I/O	SD0 data
SMIO_19	Y8					SD_D[1]	I/O	SD0 data
SMIO_20	AA8					SD_D[2]	I/O	SD0 data
SMIO_21	AA9					SD_D[3]	I/O	SD0 data
SMIO_22	Y9	NORSPI_DQ[4]	I/O			SD_D[4]	I/O	SPI NOR data / SD0 data
SMIO_23	W9	NORSPI_DQ[5]	I/O			SD_D[5]	I/O	SPI NOR data / SD0 data

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Name	Loc.	SPI NOR		SPI NAND		SD		Description
		Function	Dir	Function	Dir	Function	Dir	
SMIO_24	AA11	NORSPI_ DQ[6]	I/O			SD_D[6]	I/O	SPI NOR data / SD0 data
SMIO_25	Y11	NORSPI_ DQ[7]	I/O			SD_D[7]	I/O	SPI NOR data / SD0 data
SMIO_26	Y13	NORSPI_ CLK	O	SNAND_ SCK	O	SDIO0_CLK	O	SPI NOR / SDIO0 clock
SMIO_27	AA13	NORSPI_EN	O	SNAND_ CS_N	O	SDIO0_CMD	O	SPI NOR enable / SPI NAND chip select / SDIO0 command
SMIO_28	Y12	NORSPI_ DQ[0]	I/O	SNAND_ SI[0]	I/O	SDIO0_D[0]	I/O	SPI NOR / SPI NAND / SDIO0 data
SMIO_29	AA12	NORSPI_ DQ[1]	I/O	SNAND_ SI[1]	I/O	SDIO0_D[1]	I/O	SPI NOR / SPI NAND / SDIO0 data
SMIO_30	AA15	NORSPI_ DQ[2]	I/O	SNAND_ SI[2]	I/O	SDIO0_D[2]	I/O	SPI NOR / SPI NAND / SDIO0 data
SMIO_31	AA14	NORSPI_ DQ[3]	I/O	SNAND_ SI[3]	I/O	SDIO0_D[3]	I/O	SPI NOR / SPI NAND / SDIO0 data
SMIO_32	Y14					SDIO0_CD	I	SDIO0 card detect
SMIO_33	Y15					SDIO0_WP	O	SDIO0 write protect
SMIO_34	U5	NORSPI_ DQS	I					SPI NOR data strobe
SMIO_35	Y6					SD_RESET	O	SD0 reset

Table 3-10. Storage Media Interface Pins (SMIO) in NAND Flash and SD Modes.

Notes:

- The SD0 controller uses **SD_D[7:0]** pins. The SDIO0 controller uses the **SDIO0_D[3:0]** pins.
- SD0 requires pull-up resistors for **SD_D[7:0]** and SDIO0 requires pull-up resistors for **SDIO0_D[3:0].**

3.2.10 Pins: SSI / SPI

Name	Location	Dir	Pad Type	Description [1]
SSI0CLK	U16	I/O	CMOS	SSI0 master port bit clock
SSI0EN0	V19	O	CMOS	SSI0 master port device enable
SSI0EN1	U19	O	CMOS	SSI0 master port device enable
SSI0MISO	U17	I	CMOS	SSI0 master port data in
SSI0MOSI	U18	O	CMOS	SSI0 master port data out
SSI2CLK	N17	I/O	CMOS	SSI2 / NOR_SPI master bit clock
SSI2EN0	R18	O	CMOS	SSI2 / NOR_SPI master device enable
SSI2MISO	T19	I	CMOS	SSI2 / NOR_SPI master port data in
SSI2MOSI	T18	O	CMOS	SSI2 / NOR_SPI master port data out
SSI3CLK	P18	I/O	CMOS	SSI3 master bit clock
SSI3EN0	N18	O	CMOS	SSI3 master device enable
SSI3MISO	N19	I	CMOS	SSI3 master port data in
SSI3MOSI	P19	O	CMOS	SSI3 master port data out

Table 3-11. SSI / SPI Interface Pins.

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Note:

1. The SSI / SPI master clock speed and device-enable polarity are programmable.

3.2.11 Pins: I2C / IDC

Name	Location	Dir	Pad Type	Description
IDCSCLK	V16	I/O	CMOS	IDC slave port - clock
IDCSDATA	V17	I/O	CMOS	IDC slave port - data

Table 3-12. IDC Interface (I2C Compatible) Pins.

3.2.12 Pins: UART

Name	Location	Dir	Pad Type	Description
UART_APB_RX	W13	I	CMOS	UART APB receive
UART_APB_TX	U14	O	CMOS	UART APB transmit
UART0_AHB_CTS_N	Y18	I/O	CMOS	UART Port 0 flow control
UART0_AHB_RTS_N	U13	I/O	CMOS	UART Port 0 flow control
UART0_AHB_RX	AA18	I	CMOS	UART Port 0 AHB receive
UART0_AHB_TX	Y16	O	CMOS	UART Port 0 AHB transmit
UART1_AHB_CTS_N	Y17	I/O	CMOS	UART Port 1 flow control
UART1_AHB_RTS_N	W14	I/O	CMOS	UART Port 1 flow control
UART1_AHB_RX	V13	I	CMOS	UART Port 1 AHB receive
UART1_AHB_TX	V12	O	CMOS	UART Port 1 AHB transmit

Table 3-13. UART Interface Pins.

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3.2.13 Pins: General Purpose Input / Output (GPIO)

The table below lists the general purpose input / output (GPIO) pins on the chip. These pins have multi-function capability and are CMOS type programmable input / output.

Pin Name	GPIO	Multiplexed Function				
		First	Second	Third	Fourth	Fifth
TIMER0	0	TM11_CLK	WDT_EXT_RST_L	PWM3		SDIO0_HS_SEL
IDCSCLK	1	IDCSCLK	TM12_CLK	SSI0_EN6	VIN_STRIG0	EHCI_APP_PRT_OVCURR0
IDCSDATA	2	IDCSDATA	TM13_CLK	SSI0_EN7	VIN_STRIG1	EHCI_PRT_PWR_0
SSI0CLK	3	SSI0_SCLK	IDC0CLK	PWM4		
SSI0MOSI	4	SSI0_TXD	IDC0DATA	PWM5		
SSI0MISO	5	SSI0_RXD	IDC1CLK	PWM6		

Pin Name	GPIO	Multiplexed Function				
		First	Second	Third	Fourth	Fifth
SSI0EN0	6	SSI0_EN0	IDC1DATA	PWM7		
SSI0EN1	7	SSI0_EN1		WDT_EXT_ RST_L	SDIO0_HS_SEL	
SSI2CLK	8	SSI2_SCLK	IDC2CLK	SSI0_EN2	VSDELAY_ VSYNC_OUT_0	
SSI2MOSI	9	SSI2_TXD	IDC2DATA	SSI0_EN3	VSDELAY_ VSYNC_OUT_1	
SSI2MISO	10	SSI2_RXD	IDC3CLK	SSI0_EN4	VSDELAY_ HSYNC_OUT_0	
SSI2EN0	11	SSI2_EN0	IDC3DATA	SSI0_EN5	VSDELAY_ HSYNC_OUT_1	
SSI3CLK	12	SSI3_SCLK	IDC4CLK	SSIS_SCLK	UART3_AHB_RX	
SSI3MOSI	13	SSI3_TXD	IDC4DATA	SSIS_TXD	UART3_AHB_TX	
SSI3MISO	14	SSI3_RXD	IDC5CLK	SSIS_RXD	UART3_AHB_ CTS_N	
SSI3EN0	15	SSI3_EN0	IDC5DATA	SSIS_EN	UART3_AHB_ RTS_N	
UART_APB_RX	16	UART_APB_RX	EHCI_APP_ PRT_OVCURR0			
UART_APB_TX	17	UART_APB_TX	EHCI_PRT_ PWR_0			
PWM0	18	PWM0	VIN_STRIG0			
PWM1	19	PWM1	VIN_STRIG1			
PWM2	20	PWM2	WDT_EXT_ RST_L			
I2S_CLK	21	I2S_CLK	SSI1_SCLK			
I2S_SI_0	22	I2S_SI_0	SSI1_TXD			
I2S_SO_0	23	I2S_SO_0	SSI1_EN0			
GPIO24	24		SSI1_RXD	EHCI_APP_ PRT_OVCURR0		
GPIO25	25			EHCI_PRT_ PWR_0		
I2S_WS	26	I2S_WS				
ENET_TXEN	27	ENET_TXEN	ENET_TXEN	PWM8		
ENET_TXD_0	28	ENET_TXD_0	ENET_TXD_0	PWM9		
ENET_TXD_1	29	ENET_TXD_1	ENET_TXD_1	PWM10		
ENET_TXD_2	30		ENET_TXD_2	PWM11		
ENET_TXD_3	31		ENET_TXD_3			
ENET_RXD_0	32	ENET_RXD_0	ENET_RXD_0			
ENET_RXD_1	33	ENET_RXD_1	ENET_RXD_1			
ENET_RXD_2	34		ENET_RXD_2			
ENET_RXD_3	35		ENET_RXD_3	PWM3		
ENET_RXDV	36	ENET_RXDV	ENET_RXDV	PWM4		
ENET_MDC	37	ENET_MDC	ENET_MDC	PWM5	AHB_MDC	
ENET_MDIO	38	ENET_MDIO	ENET_MDIO	PWM6	AHB_MDIO	
ENET_PTP_PPS	39	ENET_PTP_ PPS_O	ENET_PTP_ PPS_O	PWM7		
ENET_CLK_RX	40	ENET_REF_CLK	ENET_CLK_RX	PWM8		
ENET_GTX_CLK	41	ENET_2ND_ REF_CLK	ENET_GTX_CLK	PWM9		

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Pin Name	GPIO	Multiplexed Function				
		First	Second	Third	Fourth	Fifth
ENET_EXT_OSC_CLK	42	ENET_EXT_OSC_CLK	ENET_EXT_OSC_CLK			
ENET_2ND_REF	43	ENET_2ND_REF_CLK	ENET_2ND_REF_CLK	WDT_EXT_RST_L	SSI0_EN2	
ENET1_TXEN	44	ENET1_TXEN	ENET1_TXEN			CAN2_RX
ENET1_TXD_0	45	ENET1_TXD_0	ENET1_TXD_0	SSI5_TXD		PWM11
ENET1_TXD_1	46	ENET1_TXD_1	ENET1_TXD_1	SSI5_EN0		
ENET1_TXD_2	47		ENET1_TXD_2	VSDELAY_HSYNC_OUT_0		
ENET1_TXD_3	48		ENET1_TXD_3	VSDELAY_HSYNC_OUT_1		CAN2_TX
ENET1_RXD_0	49	ENET1_RXD_0	ENET1_RXD_0	SSI5_SCLK	UART2_AHB_RX	SSIS_SCLK
ENET1_RXD_1	50	ENET1_RXD_1	ENET1_RXD_1	SSI5_RXD	UART2_AHB_TX	SSIS_TXD
ENET1_RXD_2	51	IDC5CLK	ENET1_RXD_2	VSDELAY_VSYNC_OUT_0	UART2_AHB_CTS_N	SSIS_RXD
ENET1_RXD_3	52	IDC5DATA	ENET1_RXD_3	VSDELAY_VSYNC_OUT_1	UART2_AHB_RTS_N	SSIS_EN
ENET1_RXDV	53	ENET1_RXDV	ENET1_RXDV	EXT_VSYNC_SRC_INPUT	SSI4_SCLK	CAN3_RX
ENET1_MDC	54	ENET1_MDC	ENET1_MDC	AHB_MDC1	SSI4_TXD	CAN3_TX
ENET1_MDIO	55	ENET1_MDIO	ENET1_MDIO	AHB_MDIO1	SSI4_RXD	CAN4_RX
ENET1_PTP_PPS	56	ENET1_PTP_PPS_O	ENET1_PTP_PPS_O	EXT_VSYNC_SRC_INPUT	SSI4_EN0	CAN4_TX
ENET1_CLK_RX	57	ENET1_REF_CLK	ENET1_CLK_RX		EHCI_APP_PRT_OVCURR0	CAN5_RX
ENET1_GTX_CLK	58	ENET_2ND_REF_CLK	ENET1_GTX_CLK		EHCI_PRT_PWR_0	CAN5_TX
SMIO_0	59		SNAND_SCK	NORSPI_CLK		
SMIO_1	60		SNAND_CS_N	NORSPI_EN		
SMIO_2	61		SD_CLK			
SMIO_3	62		SD_CMD			
SMIO_4	63		SD_CD			
SMIO_5	64		SD_WP			
SMIO_6	65		SNAND_SI[0]	NORSPI_DQ[0]		
SMIO_7	66		SNAND_SI[1]	NORSPI_DQ[1]		
SMIO_8	67		SNAND_SI[2]	NORSPI_DQ[2]		
SMIO_9	68		SNAND_SI[3]	NORSPI_DQ[3]		
SMIO_18	69		SD_D[0]			
SMIO_19	70		SD_D[1]			
SMIO_20	71		SD_D[2]			
SMIO_21	72		SD_D[3]			
SMIO_22	73		SD_D[4]	NORSPI_DQ[4]		
SMIO_23	74		SD_D[5]	NORSPI_DQ[5]		
SMIO_24	75		SD_D[6]	NORSPI_DQ[6]		
SMIO_25	76		SD_D[7]	NORSPI_DQ[7]		
SMIO_26	77	UART2_AHB_RX	SDIO0_CLK	SSI1_SCLK	SNAND_SCK	NORSPI_CLK
SMIO_27	78	UART2_AHB_TX	SDIO0_CMD	SSI1_TXD	SNAND_CS_N	NORSPI_EN
SMIO_28	79	UART2_AHB_CTS_N	SDIO0_D[0]	SSI1_RXD	SNAND_SI[0]	NORSPI_DQ[0]

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Pin Name	GPIO	Multiplexed Function				
		First	Second	Third	Fourth	Fifth
SMIO_29	80	UART2_AHB_RTS_N	SDIO0_D[1]	SSI1_EN0	SNAND_SI[1]	NORSPI_DQ[1]
SMIO_30	81	UART3_AHB_CTS_N	SDIO0_D[2]	SSI1_EN1	SNAND_SI[2]	NORSPI_DQ[2]
SMIO_31	82	UART3_AHB_RTS_N	SDIO0_D[3]	SSI1_EN2	SNAND_SI[3]	NORSPI_DQ[3]
SMIO_32	83	UART3_AHB_RX	SDIO0_CD	SSI1_EN3		
SMIO_33	84	UART3_AHB_TX	SDIO0_WP			
SMIO_34	85			NORSPI_DQS		
SMIO_35	86		SD_RESET			
SENSOR_RST	87		WDT_EXT_RST_L			
SHSYNC	88	IDSP_VIN0_IO-PAD_MASTER_HSYNC	IDSP_VIN1_IO-PAD_MASTER_HSYNC			
SVSYNC	89	IDSP_VIN0_IO-PAD_MASTER_VSYNC	IDSP_VIN1_IO-PAD_MASTER_VSYNC			
UART0_AHB_RX	90	UART0_AHB_RX	SSI1_SCLK	EHCI_APP_PRT_OVCURR0	TM11_CLK	PWM10
UART0_AHB_TX	91	UART0_AHB_TX	SSI1_TXD			WDT_EXT_RST_L
UART0_AHB_CTS_N	92	UART0_AHB_CTS_N	SSI1_RXD	EHCI_PRT_PWR_0		TM12_CLK
UART0_AHB_RTS_N	93	UART0_AHB_RTS_N	SSI1_EN0			TM13_CLK
UART1_AHB_RX	94	UART1_AHB_RX	TM12_CLK			
UART1_AHB_TX	95	UART1_AHB_TX				
UART1_AHB_CTS_N	96	UART1_AHB_CTS_N	IDSP_VIN1_IO-PAD_MASTER_VSYNC			
UART1_AHB_RTS_N	97	UART1_AHB_RTS_N	IDSP_VIN1_IO-PAD_MASTER_HSYNC			
CAN0_RX	98	CAN0_RX				
CAN0_TX	99	CAN0_TX				
CAN1_RX	100	CAN1_RX				
CAN1_TX	101	CAN1_TX				
SD_HS_DEL	102	SD_HS_SEL				

Table 3-14. General Purpose Input / Output Pins.

3.2.14 Pins: Real Time Clock (RTC)

Name	Location	Dir	Type	Description
XI_RTC	K4	I	XOSC	RTC clock
XO_RTC	J4	O	XOSC	RTC clock output

Table 3-15. RTC Interface Pins.

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3.2.15 Pins: Timer

Name	Location	Dir	Type	Description
TIMER0	W12	I/O	CMOS	Interval Timer 0 external clock source

Table 3-16. Timer Pins.

Note:

Frequency of the external clock source should be < 1/4 of the APB clock.

3.2.16 Pins: Pulse Width Modulator (PWM)

Name	Location	Dir	Type	Description
PWM0	W16	I/O	CMOS	Pulse Width Modulator Output 0 Used for motor control
PWM1	Y19	I/O	CMOS	Pulse Width Modulator Output 1 Used for motor control
PWM2	V14	I/O	CMOS	Pulse Width Modulator Output 2 Used for motor control

Table 3-17. PWM Pins.

3.2.17 Pins: JTAG Control

Name	Location	Dir	Pad Type	Description
JTAG_CLK	J5	I	CMOS	Clock
JTAG_RST_L	E6	I	CMOS	Reset
JTAG_TDI	G5	I	CMOS	Data in
JTAG_TDO	H5	O	CMOS	Data out
JTAG_TMS	E7	I	CMOS	Test mode select

Table 3-18. JTAG Pins.

3.2.18 Pins: Global and Test

Name	Location	Dir	Type	Description
POR_L	Y20	I	CMOS	External reset pin (active low)
TEST_MODE	T17	I	CMOS	0 - Normal mode 1 - Test mode
XIN	A9	I	XOSC	System clock (24 MHz crystal oscillator)
XOUT	B9	O	XOSC	System clock output
BG_REXT	F6	I/O	Analog	Band Gap reference resistor 10k pull low to VSS

Table 3-19. Global and Test Pins.

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3.2.19 Pins: Safety Error Status

Name	Location	Dir	Type	Description
CEHU_SAFETY_ERR_L_O	B10	O	CMOS	Safety error signal
CEHU_SAFETY_ERR_L_1	P17	O	CMOS	Safety error signal

Table 3-20. Safety Error Status Pins.

3.2.20 Pins: Power, Ground, and PLL

Name	Dir	Type	Description
VDDI	S	Power	Digital power supply
VSSI	G	Power	Digital ground (24 width)
VDDP	S	Power	IO pre-driver power
VDDO	S	Power	IO power
AVDD	S	Power	Analog core power
SD_VDDO	S	Power	SD I/O power, 3.3 V / 1.8 V
NAND_VDDO	S	Power	NAND I/O power, 3.3 V / 1.8 V
AVDD18_1	S	Power	Analog 1.8 V power
AVDD18_2	S	Power	Analog 1.8 V power
AVDD33	S	Power	Analog 3.3 V power
DDR_DVDD_VDDQ	S	Power	DDR power for main host
DDR_DVDD_IO	S	Power	DDR IO power for main host
DDRE_DVDD_VDDQ	S	Power	DDR power for ECC host
DDRE_DVDD_IO	S	Power	DDR IO power for ECC host
OTP_VPP	S	Power	OTP power
MIPI_AVDD18_IO	S	Power	1.8 V / 1.2 V MIPI I/O power

Table 3-21. Power, Ground, and PLL Pins

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4. ELECTRICAL CHARACTERISTICS

4.1 Electrical: Overview

This chapter provides details on the electrical characteristics of the CV22FS55 chip as follows:

- (Section 4.2) Electrical: Absolute Ratings
- (Section 4.3) Electrical: Recommended Operating Conditions
- (Section 4.4) Electrical: Video Signal Wave Forms and Timing
- (Section 4.5) Electrical: ENET AC Timing
- (Section 4.6) Electrical: SD Controller Timing
- (Section 4.7) Electrical: eMMC Boot Timing
- (Section 4.8) Electrical: I2S Timing
- (Section 4.9) Electrical: SSI / SPI Timing
- (Section 4.10) Electrical: Serial NAND Timing
- (Section 4.11) Electrical: I2C Timing
- (Section 4.12) Electrical: SPI NOR Timing
- (Section 4.13) Electrical: CAN Bus Timing

4.2 Electrical: Absolute Ratings

The following table gives absolute ratings for the nominal analog / digital voltages in Section 4.3.1.

Parameter	Minimum	Maximum
Analog supply voltage (3.0 V)	-0.3 V	3.6 V
Digital supply voltage (3.0 V)	-0.3 V	3.6 V
Analog supply voltage (1.8 V)	-0.3 V	1.98 V
Digital supply voltage (1.8 V)	-0.3 V	1.98 V
Analog supply voltage (0.75 V)	-0.3 V	0.88 V
Digital supply voltage (0.75 V)	-0.3 V	0.88 V
Digital I/O range (V)	-0.3 V	3.6 V
	-0.3 V	1.98 V
Analog I/O range (V)	-0.3 V	3.6 V
	-0.3 V	1.98 V
Operating junction temperature (T_J) (°C)	-40 C to +125 C	
Storage ambient temperature (T_A) (°C)	-40 C to +150 C	

Table 4-1. Absolute Ratings.

Note:

This Ambarella part will support a full range of operation at the case temperature range shown above, provided that the customer PCB design, manufacturing processes, and power supply design are equivalent to those of the Ambarella reference hardware design in terms of quality. It also is required that all other components used in the customer's system design meet this temperature range to guarantee proper system operation.

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4.3 Electrical: Recommended Operating Conditions

This section continues with recommended operating conditions for the following:

- (Section 4.3.1) Operating Conditions: Power Rails DC Characteristics
- (Section 4.3.2) Operating Conditions: Digital I/O
- (Section 4.3.3) Operating Conditions: DRAM I/O
- (Section 4.3.4) Operating Conditions: RTC Power Supply
- (Section 4.3.5) Operating Conditions: Video Sensor Input
- (Section 4.3.6) Operating Conditions: Crystal and Reference Clock Requirements

4.3.1 Operating Conditions: Power Rails DC Characteristics

Parameter	Comments	Minimum	Typical	Maximum	Ripple
RTC_VDD18		0.7 V	1.8 V	1.98 V	2%
MIPI_AVDD18_IO		1.7 V / 1.07 V	1.8 V / 1.2 V	1.9 V / 1.3 V	2%
VDDI		0.73 V	0.75 V	0.77 V	2%
VDDP		1.7 V	1.8 V	1.9 V	2%
OTP_VPP		1.7 V	1.8 V	1.9 V	2%
AVDD		0.78 V	0.8 V	0.825 V	2%
AVDD18_1		1.7 V	1.8 V	1.9 V	2%
AVDD18_2		1.7 V	1.8 V	1.9 V	2%
AVDD33		2.85 V	3.0 V	3.3 V	2%
P_SD_VDDO	3.0 V mode	2.85 V	3.0 V	3.3 V	2%
	1.8 V mode	1.7 V	1.8 V	1.9 V	2%
P_SDIO_VDDO	3.0 V mode	2.85 V	3.0 V	3.3 V	2%
	1.8 V mode	1.7 V	1.8 V	1.9 V	2%
P_NAND_VDDO	3.0 V mode	2.85 V	3.0 V	3.3 V	2%
	1.8 V mode	1.7 V	1.8 V	1.9 V	2%
VDDO	3.0 V mode	2.85 V	3.0 V	3.3 V	2%
	1.8 V mode	1.7 V	1.8 V	1.9 V	2%
DDR_DVDD_VDDQ		1.06 V	1.1 V	1.17 V	2%
DDR_DVDD_IO	LPDDR4 mode	1.06 V	1.1 V	1.17 V	2%
	LPDDR4x mode	0.57 V	0.6 V	0.65 V	2%

Table 4-2. Power Rails DC Characteristics.

Note:
The electrical details provided in this chapter are preliminary estimates. Please contact an Ambarella representative for current electrical specifications.

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4.3.2 Operating Conditions: Digital I/O

Parameter	Comments	Minimum	Typical	Maximum
VIL	Input low voltage	-0.3 V		0.3 * VDDO
VIH	Input high voltage	0.7 * VDDO		1.1 * VDDO
VOL	Output low voltage			0.2 * VDDO
VOH	Output high voltage	0.8 * VDDO		

Table 4-3. Digital I/O Characteristics (Preliminary).

Note:

- VDDO includes P_NAND_VDDO, P_SD_VDDO, P_SDIO_VDDO, as well as VDDO

4.3.3 Operating Conditions: DRAM I/O

4.3.3.1 DRAM: DC Supply Voltage Levels

Parameter	Comments	Minimum	Typical	Maximum
DDR_DVDD_VDDQ		See Section 4.3.1		
DDR_DVDD_IO		See Section 4.3.1		

Table 4-4. DRAM I/O Characteristics - DC Supply Voltage Levels (Preliminary).

4.3.3.2 DRAM: LVSTL I/O AC Specifications

Parameter	Comments	Minimum	Maximum	Application	Unit
VIH	Input HIGH voltage for normal operation	VREF+80mV		LPDDR4x/LPDDR4	V
VIL	Input LOW Voltage for normal operation		VREF-80mV	LPDDR4x/LPDDR4	V
VOH	Output HIGH voltage	0.65*DDR_DVDD_IO		LPDDR4	V
		0.8*DDR_DVDD_IO		LPDDR4x	V
VOL	Output LOW voltage		0.25*DDR_DVDD_IO	LPDDR4x/LPDDR4	V
VOH_CKE	CKE output HIGH voltage	0.75*DDR_DVDD_VDDQ		LPDDR4x/LPDDR4	V
VOL_CKE	CKE output LOW voltage		0.25*DDR_DVDD_VDDQ	LPDDR4x/LPDDR4	V

Table 4-5. DRAM I/O Characteristics - LVSTL I/O AC Specifications (Preliminary).

Notes:

- VREF range: (8.2%, 38.2%) * VDDQ
- VOH / VOL apply to DQ, DQS, command, and address
- VOH_CKE / VOL_CKE apply to CKE

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4.3.4 Operating Conditions: RTC Power Supply

Parameter	Comments	Minimum	Typical	Maximum
RTC_CP	RTC module supply	0.7 V	1.8 V	
VIH	For XI_RTC	0.4 V		1.0 V
VIL	For XI_RTC			0.2 V

Table 4-6. RTC Supply.

4.3.5 Operating Conditions: Video Sensor Input

4.3.5.1 VIN: MIPI LP Receiver DC Specification

Parameter	Comments	Minimum	Typical	Maximum
VIH	Logic 1 input voltage	880 mV		
VIL	Logic 0 input voltage, not in ULP state			550 mV
VHYST	Input hysteresis	25 mV		

Table 4-7. MIPI LP Receiver DC Specification.

4.3.5.2 VIN: MIPI LP Receiver AC Specification

Description	Minimum	Typical	Maximum
Input pulse rejection			300 V-ps
Minimum pulse width response	20 ns		
Peak interference amplitude			200 mV
Interference frequency	450 MHz		

Table 4-8. MIPI LP Receiver AC Specification.

4.3.6 Operating Conditions: Crystal and Reference Clock Requirements

4.3.6.1 Crystal and Reference Clock Requirements: 24 MHz

Description	Minimum	Typical	Maximum
Nominal frequency	N/A	24 MHz only	N/A
Frequency tolerance		± 30 PPM or better	± 80 PPM (105 C)
Load capacitance CL		10 pF	
Maximum ESR			50 Ohm
Motional capacitance		26 fF	
Shunt capacitance		7 pF	

Table 4-9. Jitter Specifications (24 MHz).

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4.3.6.2 Crystal and Reference Clock Requirements: 32.768 KHz

Description	Minimum	Typical	Maximum
Nominal frequency	N/A	32.768 KHz	N/A
Frequency tolerance		± 30 PPM or better	
Load capacitance CL		10 pF	
Maximum ESR			70 KOhm
Motional capacitance		3.4 fF	
Shunt capacitance	1 pF		

Table 4-10. Jitter Specifications (32.768 KHz).

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4.4 Electrical: Video Signal Wave Forms and Timing

4.4.1 Video Waveform: Video Input (VIN) Timing

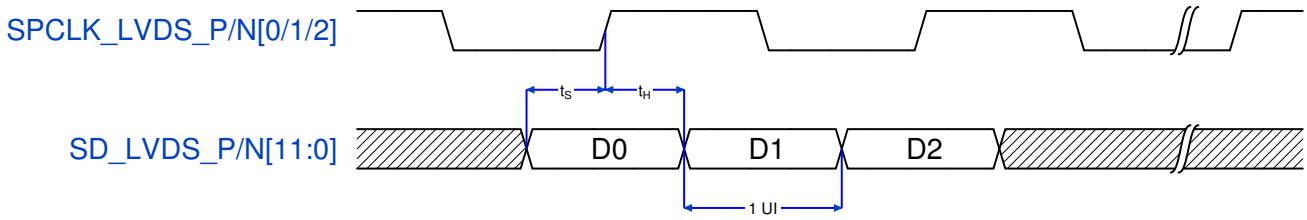


Figure 4-1. Video Input Timing.

Parameter	Setup (tS)	Hold (tH)	Comment
Data: **SPCLK_LVDS_P / N[0/1/2]** **SD_LVDS_P / N[11:0]**	75 ps	75 ps	

Table 4-11. MIPI Video Input Timing Setup / Hold.

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4.4.2 Video Output (VOUT) Timing

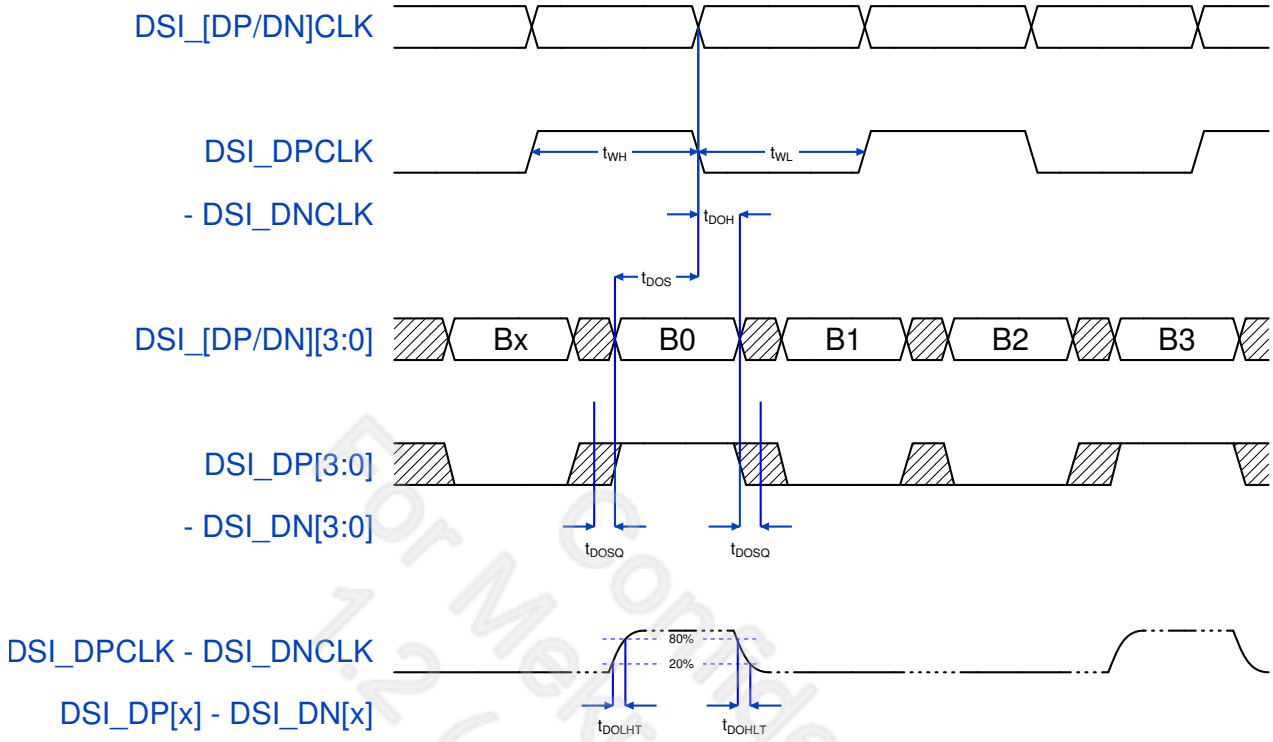


Figure 4-2. Video Output Timing.

Item	Symbol	Min.	Typ.	Max.	Unit	Comments
DO skew time (including jitter)	t_{DOSQ}			150	ps	Data rate 900 MHz DDR (1.8 Gbs)
DO setup time	t_{DOS}	100			ps	Data rate 900 MHz DDR (1.8 Gbs)
DO hold time	t_{DOH}	100			ps	Data rate 900 MHz DDR (1.8 Gbs)
DO rise time	t_{DOLHT}		200		ps	Simulated value with load capacitance (4 pF)
DO fall time	t_{DOHLT}		200		ps	Simulated value with load capacitance (4 pF)
DCK pulse width	T_{wh} T_{wl}	300			ps	Including period jitter

Table 4-12. Video Output Timing Values: MIPI DSI.

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4.5 Electrical: ENET AC Timing

4.5.1 RGMII AC Timing

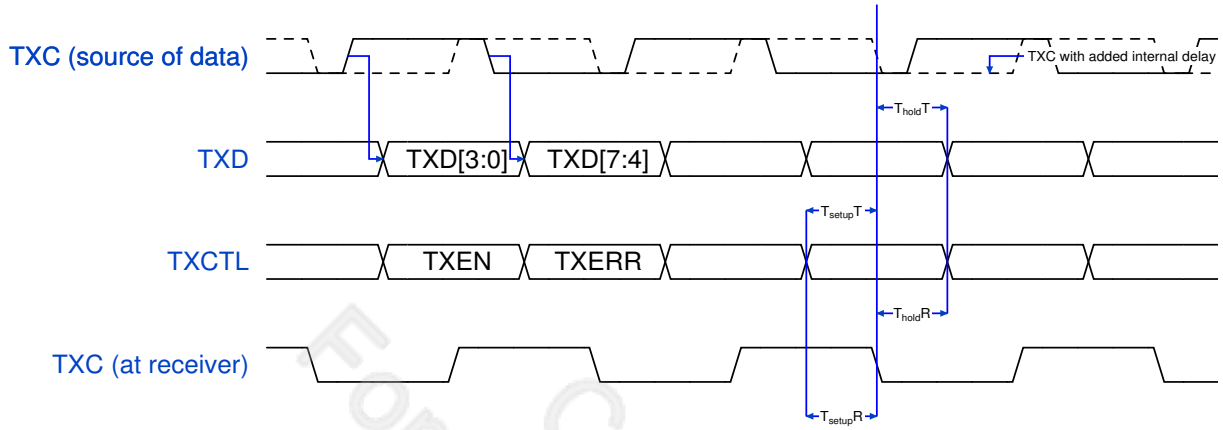


Figure 4-3. RGMII TX AC Timing.

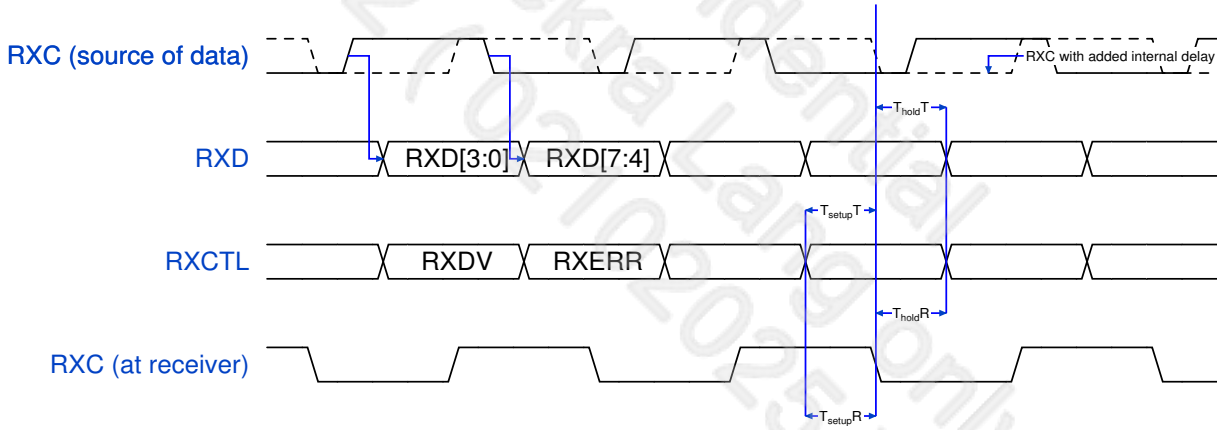


Figure 4-4. RGMII RX AC Timing.

Parameter	Symbol	Min	Typ	Max	Unit
Clock cycle duration	T_{cyc}		8		ns
Data to clock output setup (at Transmitter - integrated delay)	$T_{setup}T$	1.2	2.0		ns
Clock to data output hold (at Transmitter - integrated delay)	$T_{hold}T$	1.2	2.0		ns
Data to clock input setup (at Receiver - integrated delay)	$T_{setup}R$	1.0	2.0		ns
Clock to data input hold (at Receiver - integrated delay)	$T_{hold}R$	1.0	2.0		ns
Rise / fall time (20% - 80%)	T_r/T_f			0.75	ns

Table 4-13. RGMII AC Timing Parameters.

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4.5.2 RMII AC Timing

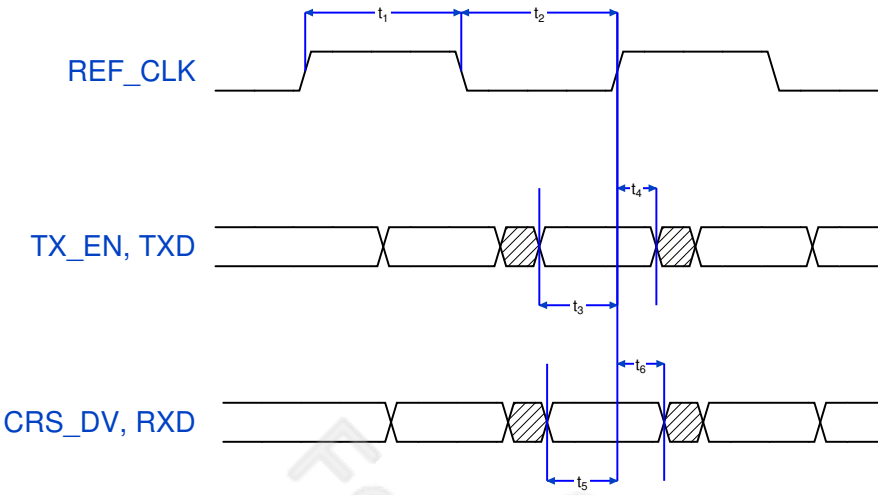


Figure 4-5. RMII AC Timing.

Parameter	Symbol	Min	Typ	Max	Unit
REF_CLK frequency			50		MHz
REF_CLK high pulse width	t_1	7		13	ns
REF_CLK low-pulse width	t_2	7		13	ns
TXEN, TXD[1:0] setup	t_3	4			ns
TXEN, TXD[1:0] hold	t_4	2			ns
CRS_DV, RXD[1:0] setup	t_5	4			ns
CRS_DV, RXD[1:0] hold	t_6	2			ns

Table 4-14. RMII AC Timing Parameters.

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4.5.3 MDC / MDIO Timing

The period of management data clock (MDC) is derived from **gclk_core**. Per the setting of **cr** register in **MAC_R4_RMII_address** register of Ethernet controller, it can be in the range of 1.25 and 2.8 MHz, or even higher if necessary.

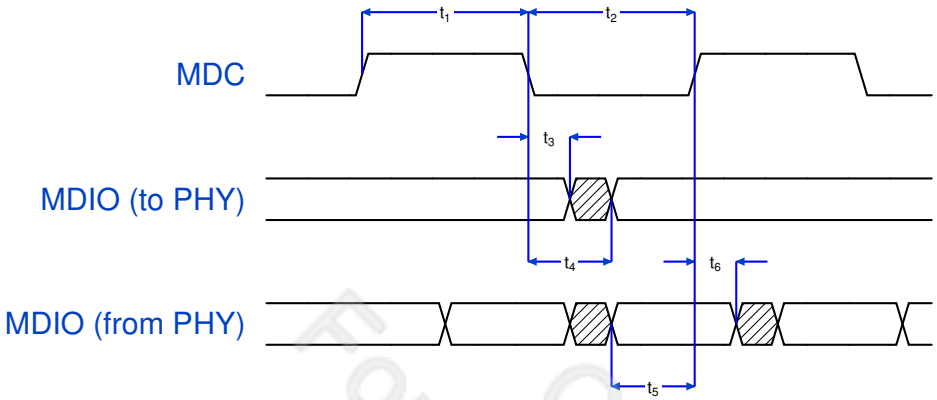


Figure 4-6. MDC / MDIO Timing.

Parameter	Symbol	Min	Typ	Max	Unit
MDC pulse width high	t_1	40%		60%	MDC period
MDC pulse width low	t_2	40%		60%	MDC period
MDIO output hold time	t_3	0			ns
MDIO time to valid output	t_4			5	ns
MDIO input setup time	t_5	18			ns
MDIO input hold time	t_6	0			ns

Table 4-15. MDC / MDIO Timing Parameters.

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4.6 Electrical: SD Controller Timing

There are two voltages at which the SD card is operated.

- (Section 4.6.1) SD Controller Timing: 3.3 V
- (Section 4.6.2) SD Controller Timing: 1.8 V

4.6.1 SD Controller Timing: 3.3 V

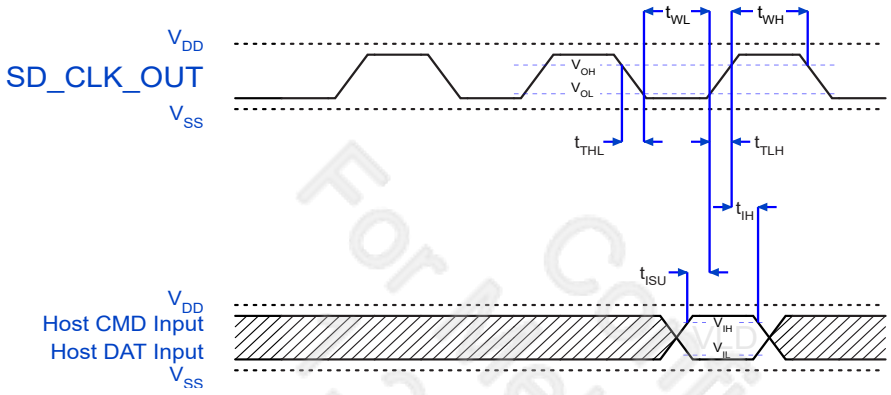


Figure 4-7. SD Host Input Timing.

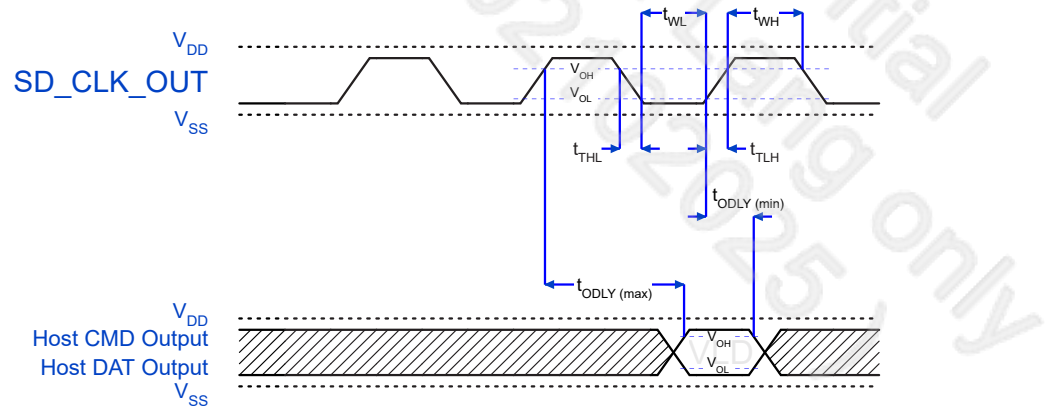


Figure 4-8. SD Host Output Timing.

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Parameter	Symbol	Min	Max	Unit	Comment
Clock CLK: All values are referred to as min (VIH) and max (VIL)					
Clock frequency: data transfer mode	f_{PP}	0	50	MHz	C_{CARD} ≤ 10 pF (1 card)
Clock frequency: identification mode	f_{OD}	0/100	400	kHz	C_{CARD} ≤ 10 pF (1 card)
Clock low time	t_{WL}	7		ns	C_{CARD} ≤ 10 pF (1 card)
Clock high time	t_{WH}	7		ns	C_{CARD} ≤ 10 pF (1 card)
Clock rise time	t_{TLH}		3	ns	C_{CARD} ≤ 10 pF (1 card)
Clock fall time	t_{THL}		3	ns	C_{CARD} ≤ 10 pF (1 card)
Inputs CMD, DAT: Referenced to CLK					
Input set-up time	t_{ISU}	5.83		ns	C_{CARD} ≤ 10 pF (1 card)
Input hold time	t_{IH}	1.5		ns	C_{CARD} ≤ 10 pF (1 card)
Outputs CMD, DAT: Referenced to CLK at 50 MHz					
Output delay time	t_{ODLY}	10.8	13.4	ns	C_L = 30 pF

Table 4-16. SD Controller Timing Parameters.

Note:

The starting point of t_{ODLY} is from the rising edge of SD_CLK_OUT and is clock-frequency-dependent, assuming that the clock period of SD_CLK_OUT is T_{PP}, t_{ODLY} has the range of $0.5*T_{PP}+0.8$ and $0.5*T_{PP}+3.4$.

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4.6.2 SD Controller Timing: 1.8 V

The card input setup time and hold time are measured at V_{IL} (max.) and V_{IH} (min.). The timing parameter, clock threshold (V_{CT}), is used to indicate clock reference point and is defined as 0.975 V.

Parameter	Symbol	Min	Max	Unit	Comment
Clock CLK: All values are referred to as min (VIH) and max (VIL)					
Clock period	t_{CLK}	10	-	ns	100 MHz (Max). Between rising edge. V_{CT}=0.975 V
Clock rising / failing time	t_{CR}, t_{CF}	-	0x2 * t_{CLK}	ns	C_{CARD} = 10 pF
Clock duty	-	30	70	%	
Host Inputs CMD, DAT: Referenced to CLK					
Input set-up time	t_{IS}	5.83	-	ns	C_{CARD} = 10 pF, V_{CT}=0.975V
Input hold time	t_{IH}	1.5	-	ns	C_{CARD} = 5 pF, V_{CT}=0.975V
Host Outputs CMD, DAT: Referenced to CLK at 50 MHz					
Output delay time	t_{ODLY}	-	3.4	ns	C_L = 30 pF
Output hold time	t_{OHLD}	0.8	-	ns	C_L = 30 pF

Table 4-17. SD Controller Timing Parameters in SDR50 Mode.

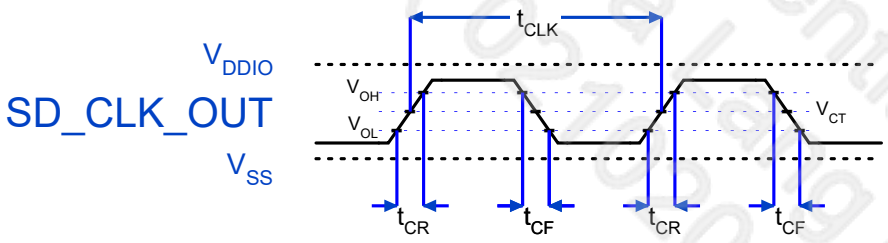


Figure 4-9. Clock Signal Timing.

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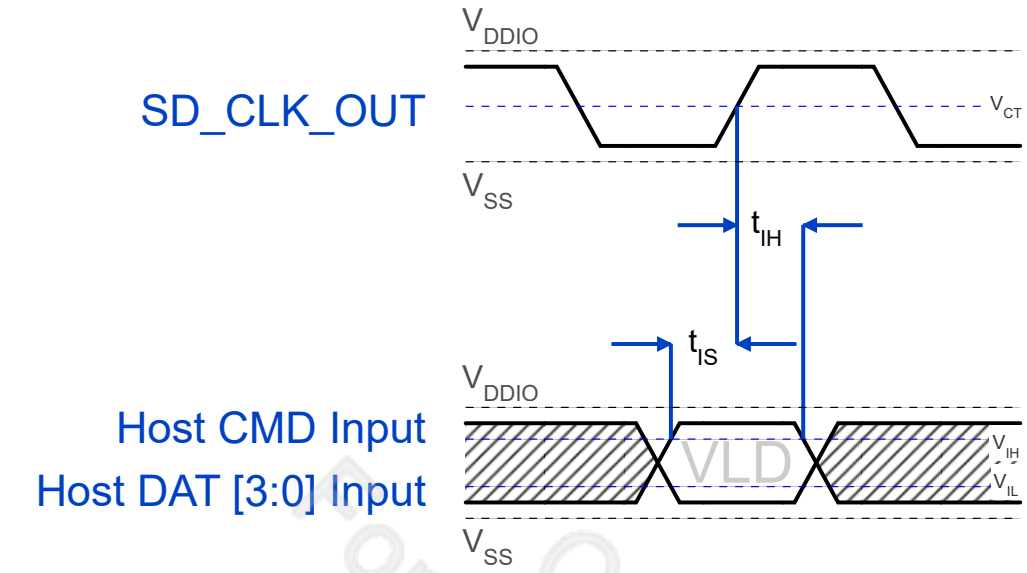


Figure 4-10. SD Host Input Timing.

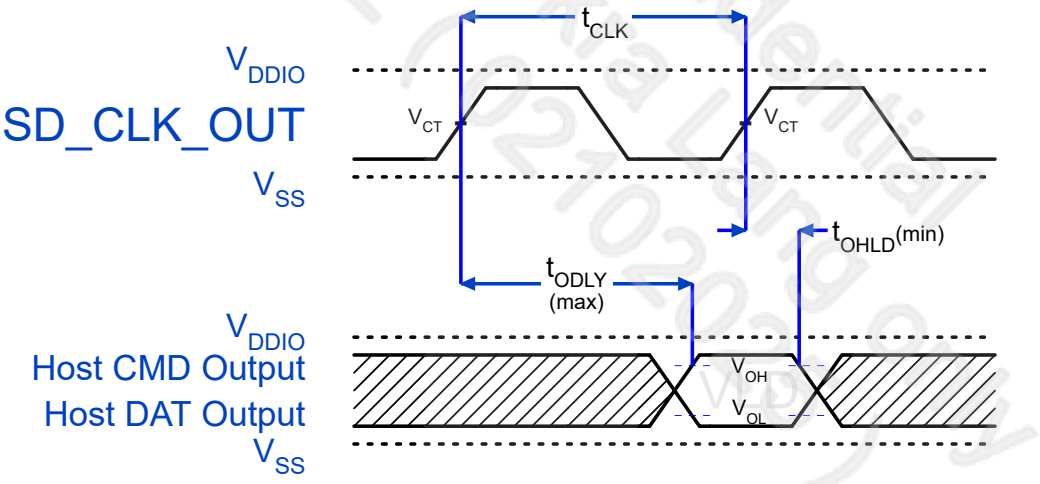


Figure 4-11. SD Host Output Timing.

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4.7 Electrical: eMMC Boot Timing

To successfully boot from eMMC, the eMMC device should return boot data with the following timing constraints.

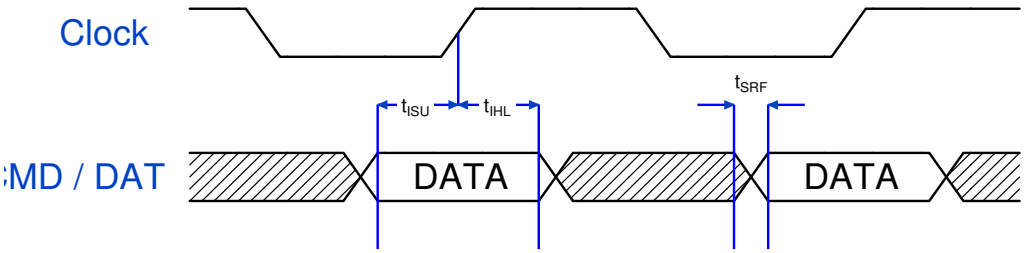


Figure 4-12. eMMC Boot Timing Diagram.

Parameter	Symbol	Minimum	Maximum
Host CMD / DAT input timing			
Input setup time	t_{ISU}	5.83 ns	
Input hold time	t_{IHL}	1.5 ns	
Signal rise time	t_{SRF}		3 ns
Signal Fall Time			3 ns

Table 4-18. eMMC Boot Timing.

Note:

CMD / DAT input rise and fall time are measured by VIL (max) and VIH (min).

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4.8 Electrical: I2S Timing

4.8.1 I2S Master Timing

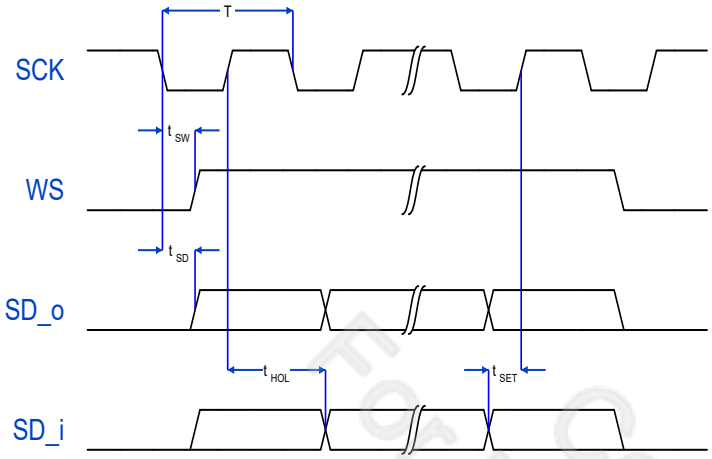


Figure 4-13. Timing Diagram for I2S Master (tsp=1, rsp=0)[1].

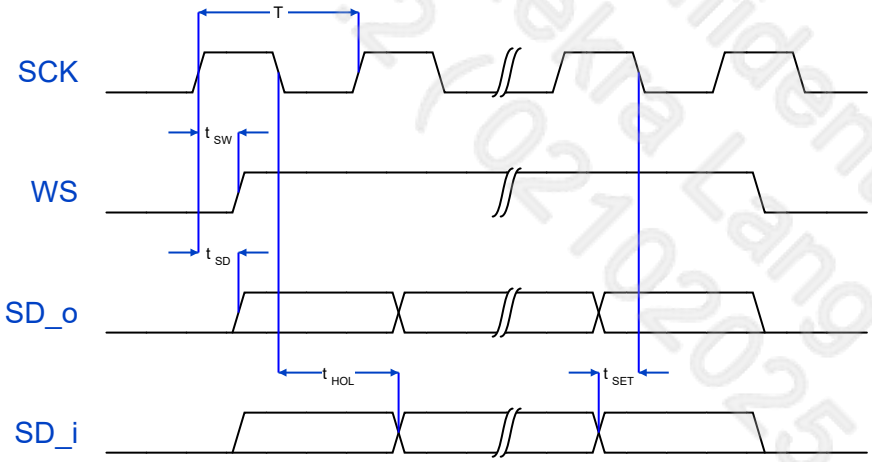


Figure 4-14. Timing Diagram for I2S Master (tsp=0, rsp=1).

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Clock drive edge to word select change	t_{SW}	-	5	ns
Clock drive edge to SD_o change	t_{SD}	-	5	ns
SD_i setup time	t_{SET}	16	-	ns
SD_i hold time	t_{HOL}	0	-	ns

Table 4-19. Timing Parameters for I2S Master.

Note:

[1] Configuration Register at offset 0x20 bit [6:5]

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4.8.2 I2S Slave Timing

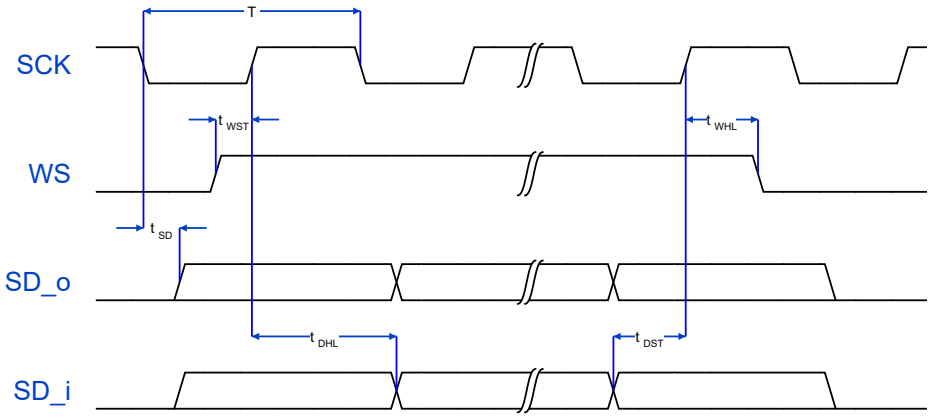


Figure 4-15. Timing Diagram for I2S Slave (tsp=1, rsp=0)[1].

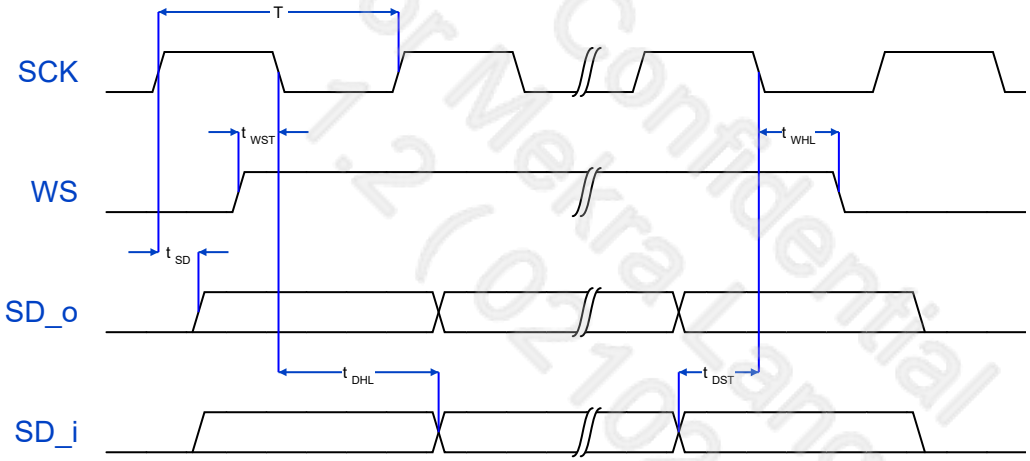


Figure 4-16. Timing Diagram for I2S Slave (tsp=0, rsp=1).

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Word select setup time	t_{WST}	8	-	ns
Word select hold time	t_{WHL}	0	-	ns
Clock drive edge to SD_o change	t_{SD}	-	13	ns
SD_i setup time	t_{DST}	7	-	ns
SD_i hold time	t_{DHL}	0	-	ns

Table 4-20. Timing Parameters for I2S Slave.

Note:

[1] Configuration Register at offset 0x20 bit [6:5]

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4.9 Electrical: SSI / SPI Timing

4.9.1 SSI / SPI Master Timing

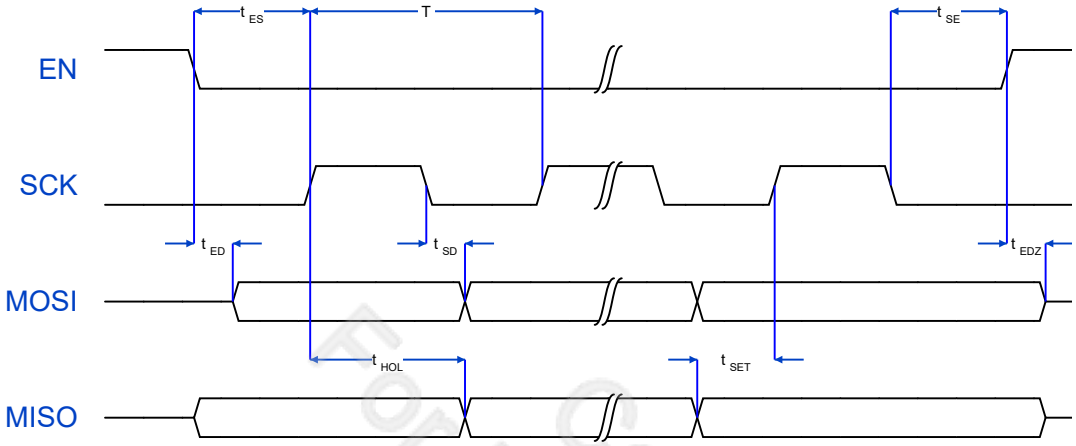


Figure 4-17. Timing Diagram for SSI / SPI Master (SCPOL = 0, SCPH = 0)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Slave enable low to clock high	t_{ES}	T-3	-	ns
Clock low to slave enable high	t_{SE}	0.5T-3	-	ns
MISO setup time	t_{SET}	12-rx_margin[2]	-	ns
MISO hold time	t_{HOL}	0	-	ns
Slave enable to MOSI valid from high impedance	t_{ED}	-	0.5T+3	ns
Clock low to MOSI valid	t_{SD}	-	3	ns
Slave enable high to MOSI high impedance	t_{EDZ}	-	8	ns

Table 4-21. Timing Parameters for SSI / SPI Master (SCPOL = 0, SCPH = 0).

Notes:

[1] Configuration register at offset 0x00 bits [7:6]
[2] Configuration register at offset 0x00 bits [25:22]

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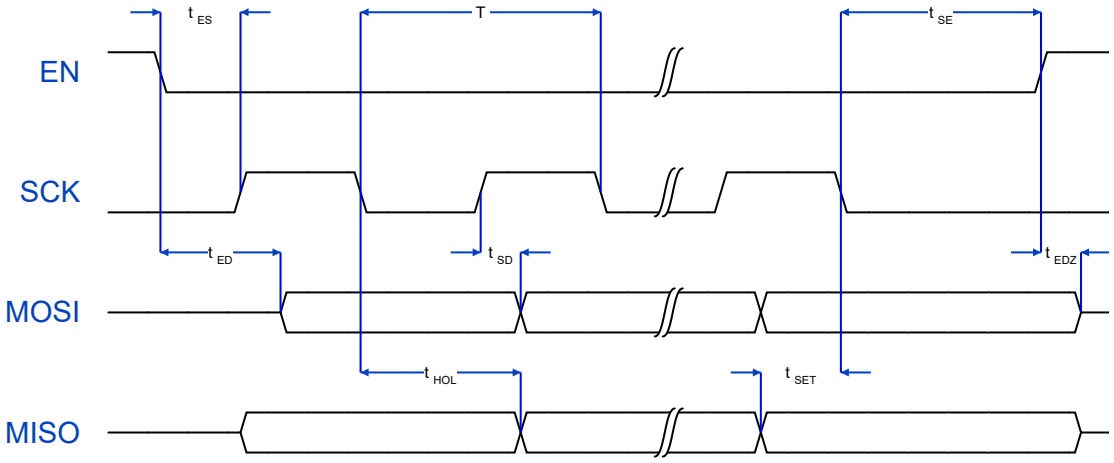


Figure 4-18. Timing Diagram for SPI Master (SCPOL = 0, SCPH = 1)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Slave enable low to clock high	t_{ES}	0.5T-3	-	ns
Clock low to slave enable high	t_{SE}	T-3	-	ns
MISO setup time	t_{SET}	12-rx_margin[2]	-	ns
MISO hold time	t_{HOL}	0	-	ns
Slave enable to MOSI valid from high impedance	t_{ED}	-	0.5T+3	ns
Clock high to MOSI valid	t_{SD}	-	3	ns
Slave enable high to MOSI high impedance	t_{EDZ}	-	8	ns

Table 4-22. Timing Parameters for SSI / SPI Master (SCPOL = 0, SCPH = 1).

Notes:

[1] Configuration register at offset 0x00 bits [7:6]
[2] Configuration register at offset 0x00 bits [25:22]

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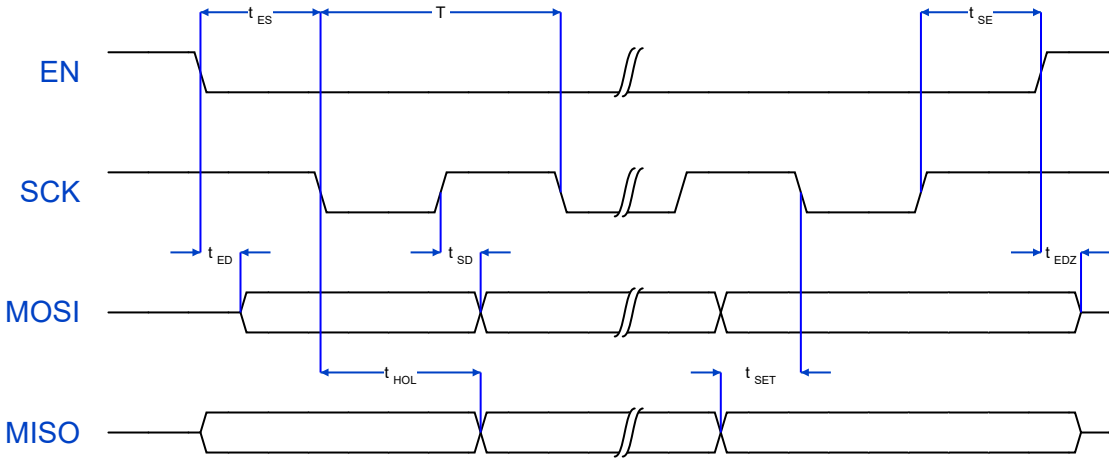


Figure 4-19. Timing Diagram for SPI Master (SCPOL = 1, SCPH = 0)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Slave enable low to clock low	t_{ES}	T-3	-	ns
Clock high to slave enable high	t_{SE}	0.5T-3	-	ns
MISO setup time	t_{SET}	12-rx_margin[2]	-	ns
MISO hold time	t_{HOL}	0	-	ns
Slave enable to MOSI valid from high impedance	t_{ED}	-	0.5T+3	ns
Clock high to MOSI valid	t_{SD}	-	3	ns
Slave enable high to MOSI high impedance	t_{EDZ}	-	8	ns

Table 4-23. Timing Parameters for SSI / SPI Master (SCPOL = 1, SCPH = 0).

Notes:

[1] Configuration register at offset 0x00 bits [7:6]
[2] Configuration register at offset 0x00 bits [25:22]

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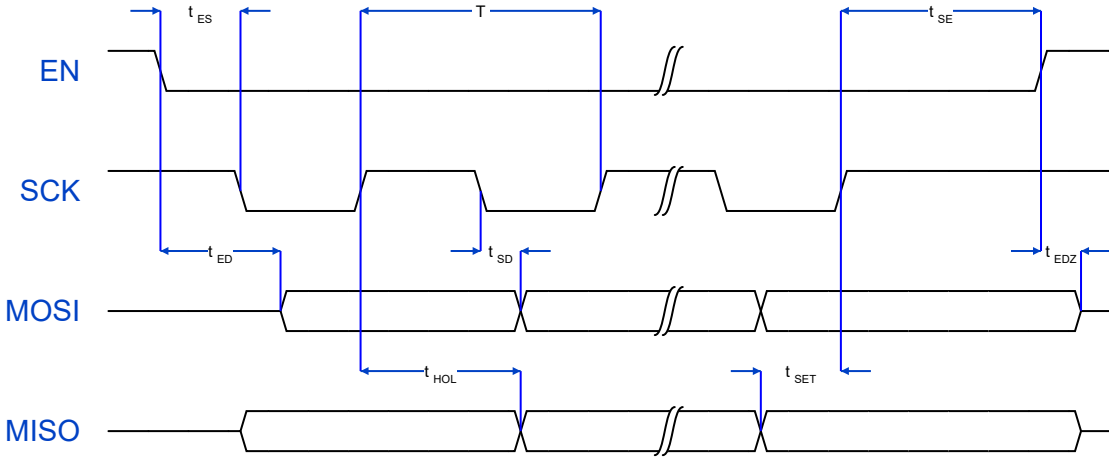


Figure 4-20. Timing Diagram for SPI Master (SCPOL = 1, SCPH = 1)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	50	-	ns
Slave enable low to clock low	t_{ES}	0.5T-3	-	ns
Clock high to slave enable high	t_{SE}	T-3	-	ns
MISO setup time	t_{SET}	12-rx_margin[2]	-	ns
MISO hold time	t_{HOL}	0	-	ns
Slave enable to MOSI valid from high impedance	t_{ED}	-	0.5T+3	ns
Clock low to MOSI valid	t_{SD}	-	3	ns
Slave enable high to MOSI high impedance	t_{EDZ}	-	8	ns

Table 4-24. Timing Parameters for SSI / SPI Master (SCPOL = 1, SCPH = 1).

Notes:

[1] Configuration register at offset 0x00 bits [7:6]
[2] Configuration register at offset 0x00 bits [25:22]

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4.9.2 SSI / SPI Slave Timing

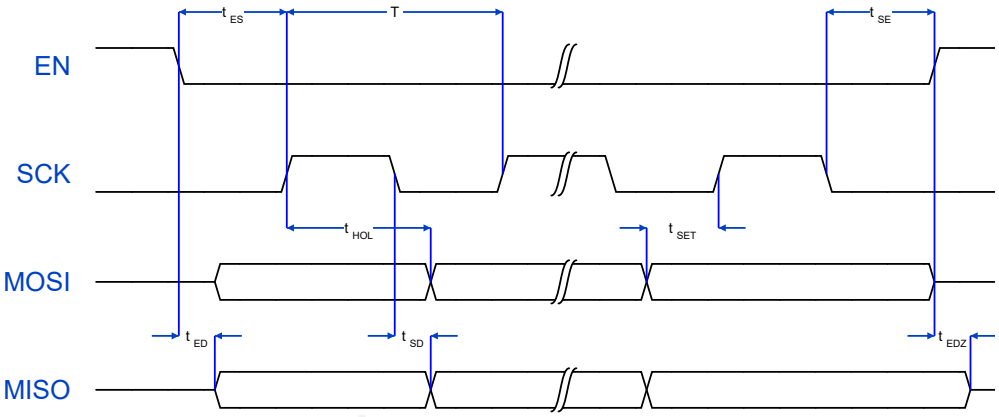


Figure 4-21. Timing Diagram for SSI / SPI Slave (SCPOL = 0, SCPH = 0)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	100	-	ns
Slave enable low to clock high	t_{ES}	0.5T+1	-	ns
Clock low to slave enable high	t_{SE}	1	-	ns
MOSI setup time	t_{SET}	0	-	ns
MOSO hold time	t_{HOL}	3 *T_{SSI_CLK}[2]+1	-	ns
Slave enable to MISO valid from high impedance	t_{ED}	-	18+6 *T_{SSI_CLK}	ns
Clock low to MISO valid	t_{SD}	-	18+4 *T_{SSI_CLK}	ns
Slave enable high to MISO high impedance	t_{EDZ}	-	13+4 *T_{SSI_CLK}	ns

Table 4-25. Timing Parameters for SSI / SPI Slave (SCPOL = 0, SCPH = 0).

Notes:

[1] Configuration register at offset 0x00 bit [7:6]
[2] Internal SPI clock period

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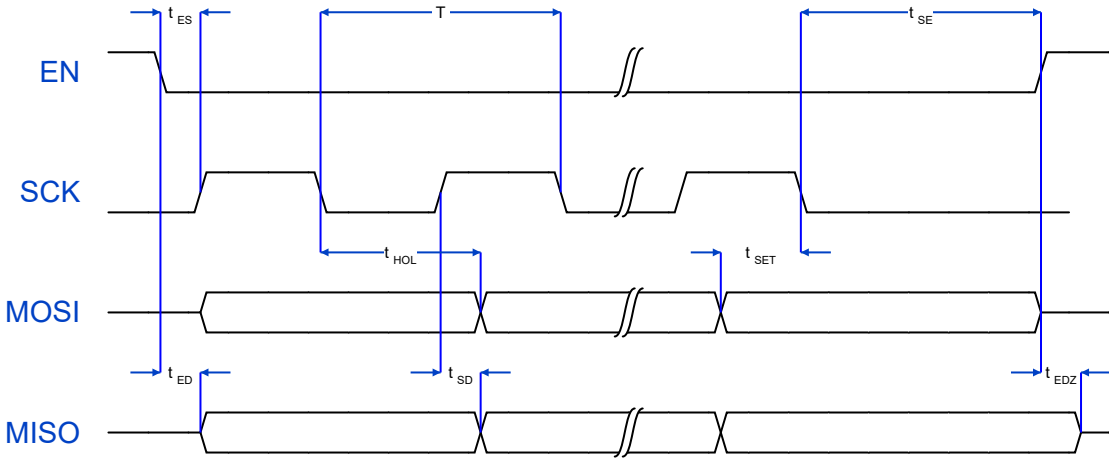


Figure 4-22. Timing Diagram for SSI / SPI Slave (SCPOL = 0, SCPH = 1)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	100	-	ns
Slave enable low to clock high	t_{ES}	4 *T_{SSI_CLK}[2]+1	-	ns
Clock low to slave enable high	t_{SE}	5 *T_{SSI_CLK}+1	-	ns
MOSI setup time	t_{SET}	0	-	ns
MOSO hold time	t_{HOL}	3 *T_{SSI_CLK}+1	-	ns
Slave enable to MISO valid from high impedance	t_{ED}	-	18+6 *T_{SSI_CLK}	ns
Clock high to MISO valid	t_{SD}	-	18+4 *T_{SSI_CLK}	ns
Slave enable high to MISO high impedance	t_{EDZ}	-	13+4 *T_{SSI_CLK}	ns

Table 4-26. Timing Parameters for SSI / SPI Slave (SCPOL = 0, SCPH = 1).

Notes:

[1] Configuration register at offset 0x00 bit [7:6]
[2] Internal SPI clock period

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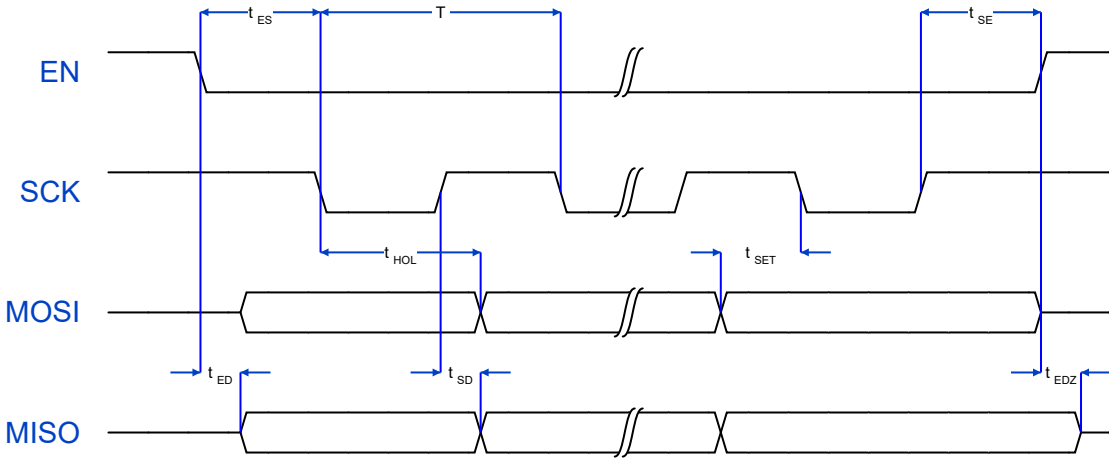


Figure 4-23. Timing Diagram for SSI / SPI Slave (SCPOL = 1, SCPH = 0)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	100	-	ns
Slave enable low to clock low	t_{ES}	0.5T+1	-	ns
Clock high to slave enable high	t_{SE}	1	-	ns
MOSI setup time	t_{SET}	0	-	ns
MOSO hold time	t_{HOL}	3 *T_{SSI_CLK}[2]+1	-	ns
Slave enable to MISO valid from high impedance	t_{ED}	-	18+6 *T_{SSI_CLK}	ns
Clock high to MISO valid	t_{SD}	-	18+4 *T_{SSI_CLK}	ns
Slave enable high to MISO high impedance	t_{EDZ}	-	13+4 *T_{SSI_CLK}	ns

Table 4-27. Timing Parameters for SSI / SPI Slave (SCPOL = 1, SCPH = 0).

Notes:

[1] Configuration register at offset 0x00 bit [7:6]
[2] Internal SPI clock period

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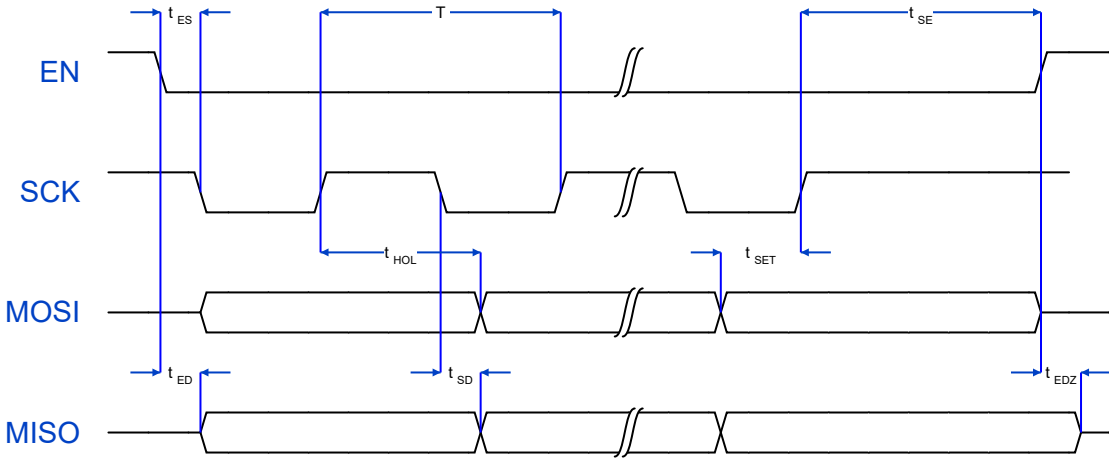


Figure 4-24. Timing Diagram for SSI / SPI Slave (SCPOL = 1, SCPH = 1)[1].

Parameter	Symbol	Min	Max	Unit
Clock period	T	100	-	ns
Slave enable low to clock low	t_{ES}	4 *T_{SSI_CLK}[2]+1	-	ns
Clock high to slave enable high	t_{SE}	5 *T_{SSI_CLK}+1	-	ns
MOSI setup time	t_{SET}	0	-	ns
MOSO hold time	t_{HOL}	3 *T_{SSI_CLK}+1	-	ns
Slave enable to MISO valid from high impedance	t_{ED}	-	18+6 *T_{SSI_CLK}	ns
Clock low to MISO valid	t_{SD}	-	18+4 *T_{SSI_CLK}	ns
Slave enable high to MISO high impedance	t_{EDZ}	-	13+4 *T_{SSI_CLK}	ns

Table 4-28. Timing Parameters for SSI / SPI Slave (SCPOL = 1, SCPH = 1).

Notes:

[1] Configuration register at offset 0x00 bit [7:6]
[2] Internal SPI clock period

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4.10 Electrical: Serial NAND Timing

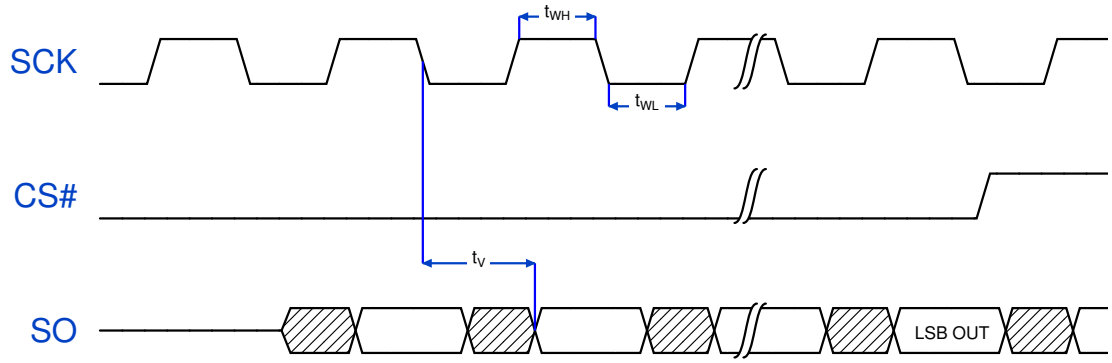


Figure 4-25. Timing for Serial NAND Input.

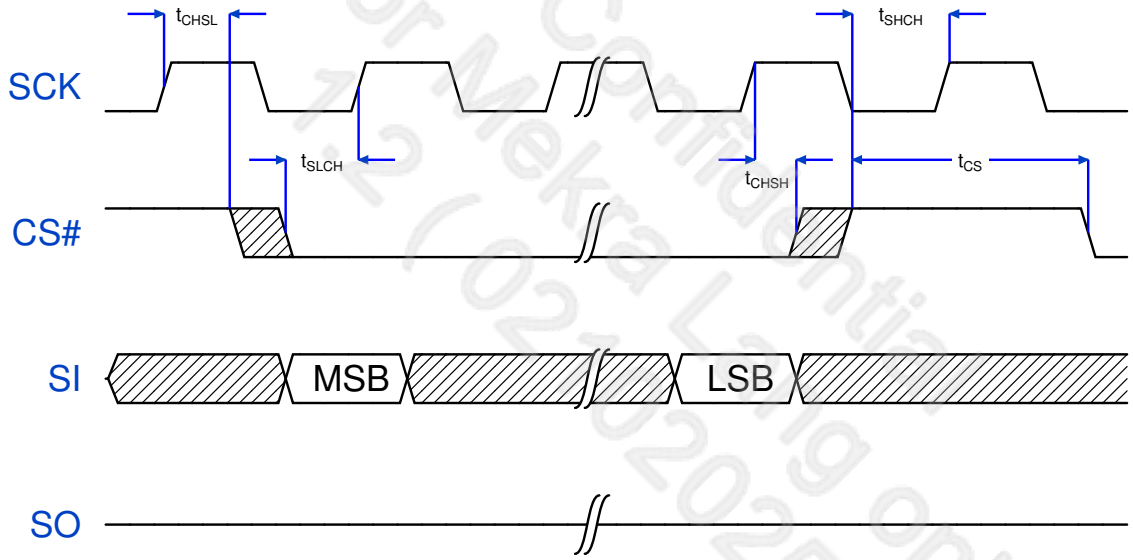


Figure 4-26. Timing for Serial NAND Output.

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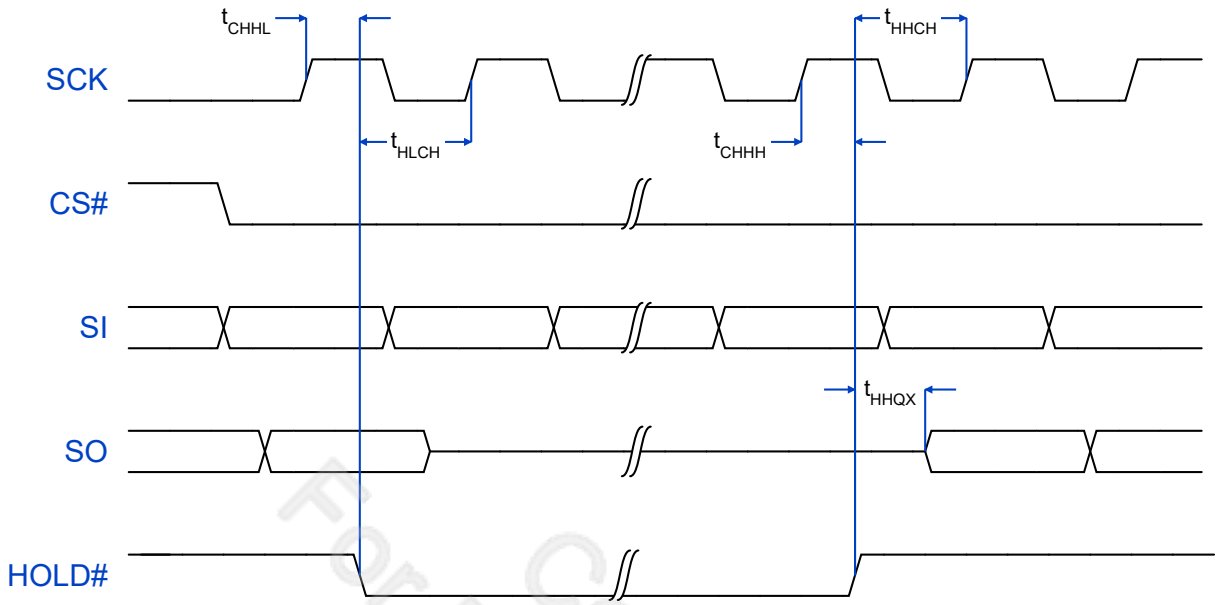


Figure 4-27. Timing for Serial NAND Hold.

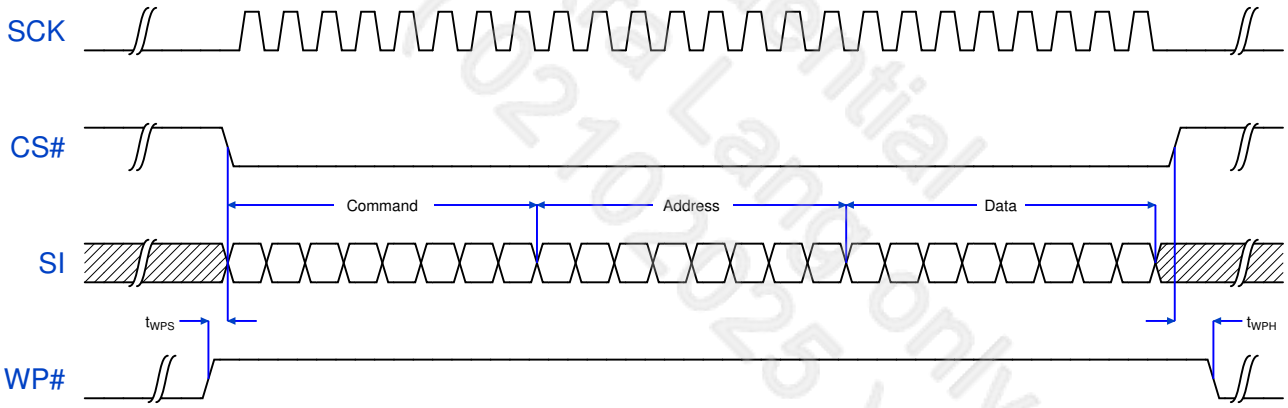


Figure 4-28. Timing for Serial NAND WP.

Parameter	Symbol	Standard Mode		Unit
		Min	Max	
Clock frequency	fc	-	133	MHz
SCK high time	t_{WH}	(snd_tclh_cnum+1)*T_{nand_pll}		ns
SCK low time	t_{WL}	(snd_tcll_cnum+1)*T_{nand_pll}		ns
CS# deselect time	t_{CS}	(snd_tcs_cnum+1)*T_{nand_pll}		ns
SCK low to read-data valid	t_V	T_{HO}	T_{HO} + 6	ns

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Parameter	Symbol	Standard Mode		Unit
		Min	Max	
SCK rising edge to CS# falling edge	t_{CHSL}	(snd_tchsl_cnum+1)*T_{nand_pll}+0.2	-	ns
CS# falling edge to SCK rising edge	t_{SLCH}	(snd_tslch_cnum+1)*T_{nand_pll}+0.2	-	ns
SCK rising edge to CS# rising edge	t_{CHSH}	(snd_tchsh_cnum+1)*T_{nand_pll}+0.2	-	ns
CS# rising edge to SCK rising edge	t_{SHCH}	(snd_tshch_cnum+1)*T_{nand_pll}+0.2	-	ns
HOLD# rising edge to read-data valid	t_{HHQX}	T_{HO}	T_{HO} + 6	ns
WP# rising edge to CS# falling edge	t_{WPS}	(snd_twps_cnum+1)*T_{nand_pll}±0.5		ns
CS# rising edge to WP# falling edge	t_{WPH}	(snd_twph_cnum+1)*T_{nand_pll}±0.5		ns
HOLD# falling edge to SCK rising edge	t_{HLCH}	(snd_thlch_cnum+1)*T_{nand_pll}+0.2	-	ns
HOLD# rising edge to SCK rising edge	t_{HHCH}	(snd_thhch_cnum+1)*T_{nand_pll}+0.2	-	ns
SCK rising edge to HOLD# falling edge	t_{CHHL}	(snd_tchhl_cnum+1)*T_{nand_pll}+0.2	-	ns
SCK rising edge to HOLD# rising edge	t_{CHHH}	(snd_tchhh_cnum+1)*T_{nand_pll}+0.2	-	ns

Table 4-29. Timing Parameters for Serial NAND.

Notes:

- T_{nand_pll} stands for NAND PLL clock period.
- T_{HO} stands for the output hold time of the NAND device.
- snd_t*_cnum stands for configuration registers.

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4.11 Electrical: I2C Timing

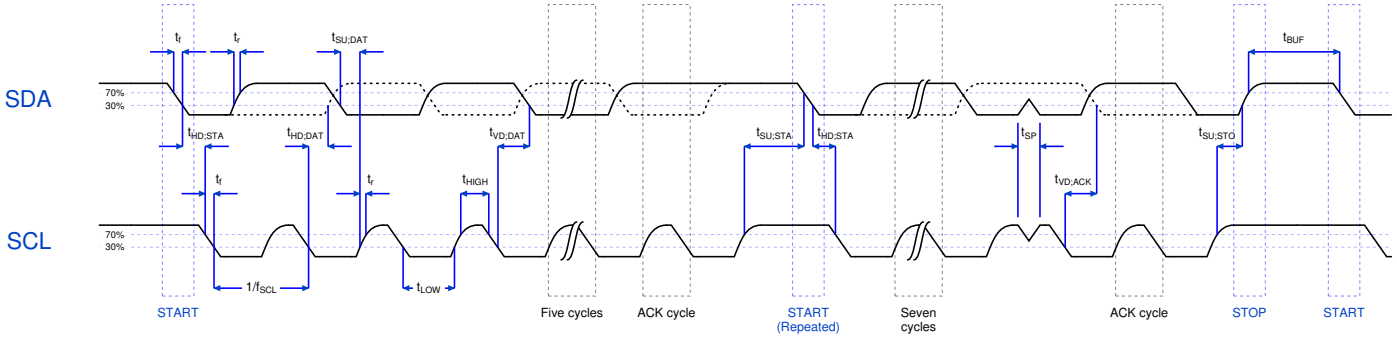


Figure 4-29. Definition of Timing for F/S-Mode Devices on the I2C-Bus.

Parameter	Symbol	Standard Mode		Fast Mode		Unit
		Min	Max	Min	Max	
SCL cycle time	f_{SCL}	0	100	0	400	KHz
Set-up time START condition	$t_{SU;STA}$	4.7		0.6		us
Hold time START condition	$t_{HD;STA}$	4.0		0.6		us
Low period of SCL	t_{LOW}	4.7		1.3		us
High period of SCL	t_{HIGH}	4.0		0.6		us
Data set-up time	$t_{SU;DAT}$	250		100		ns
Data hold time	$t_{HD;DAT}$	0	3.45	0	0.9	us
Rise time	t_{RISE}		1000	20	300	ns
Fall time	t_{FALL}		300	20 x (V_{DD} / 5.5 V)	300	ns
Set-up time STOP condition	$t_{SU;STO}$	4.0		0.6		us
Bus free time between STOP and START condition	t_{BUF}	4.7		1.3		us
Capacitive load for each bus line	C_b		400		400	pF

Table 4-30. I2C Characteristic of SDA and SCL for F/S-Mode.

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4.12 Electrical: SPI NOR Timing

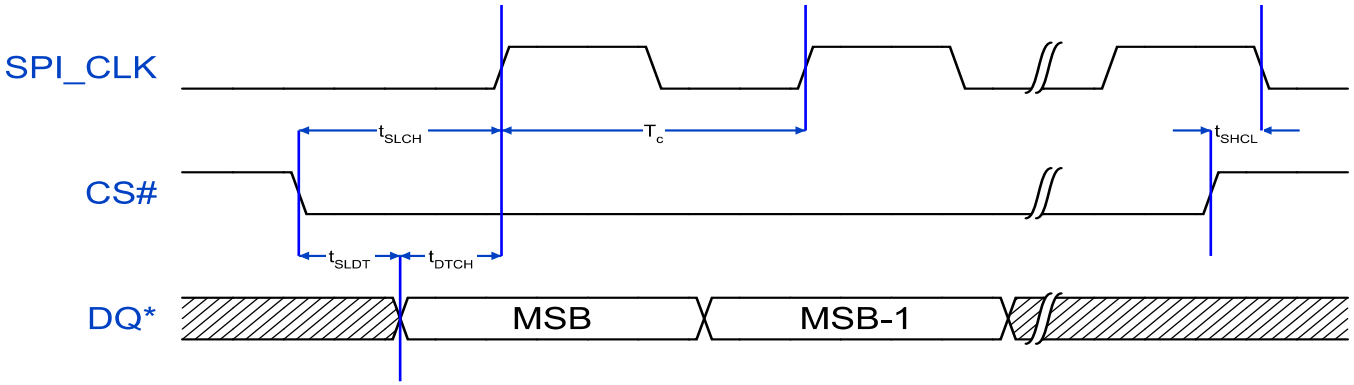


Figure 4-30. Timing for SPI NOR.

Notes:

- reg_clkdivider: 0x8 [17:10] clock divider
- fref: reference clock frequency (GCLK_SSI3)
- Tref: reference clock period (GCLK_SSI3)

Parameter	Symbol	Min	Max	Unit	Note
SPI_CLOCK frequency	f_c	-	100	MHz	1
SPI_CLOCK period	T_C	reg_clkdivider*Tref		ns	
CS# falling edge to SPI_ CLOCK rising edge	t_{SLCH}	$(T_C/2)$ + Tref - 2	$(T_C/2)$ + Tref + 2	ns	
DQ* transition to SPI_ CLOCK rising edge (STR)	t_{DTCH}	$(T_C/2)$ - 2	$(T_C/2)$ + 2	ns	
DQ* transition to SPI_ CLOCK rising edge (DTR)		$(T_C/4)$ - 2	$(T_C/4)$ + 2	ns	
CS# falling edge to DQ* transition	t_{SLDT}	T_{ref} - 2	T_{ref} + 2	ns	
CS# rising edge to SPI_CLK falling edge	t_{SHCL}	-2	$(2*T_{ref})$ + 2	ns	

Table 4-31. Timing Parameters for SPI NOR.

Note:

1. Max fc is 100 MHz when fref = 400 MHz and reg_clkdivider = 4

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4.13 Electrical: CAN Bus Timing

Parameter	Symbol	Min	Max	Unit	Notes
CANC frequency (Nominal bit-rate)	Fc(N)	gclk_can / (((tseg1_m1 + 1) + (tseg2_m1 + 1) + 1) * (tq_pre_m1+ 1))		MHz	1
CANC frequency (Data bit-rate)	Fc(D)	gclk_can / (((tseg1_ fd_m1 + 1) + (tseg2_ fd_m1 + 1) + 1) * (tq_pre_fd_m1+ 1))		MHz	2
CAN TX RX round trip delay inside chip (LOW)	tTLRL	-	13	ns	3, 4
CAN TX RX round trip delay inside chip (HIGH)	tHRH	-	13	ns	3, 4

Table 4-32. Timing Parameters for CAN Bus.

Notes:

1. Maximum Fc(N) = 1 MHz depending on transceiver speed. Typical CAN transceiver is up to 1 Mbps
2. Maximum Fc(D) = 8 MHz depending on transceiver speed. Typical CAN-FD transceiver is up to 8 Mbps
3. CAN TX RX round trip delay inside chip = delay from can*_tx register to can*_tx output pad + delay from can*_rx input pad to can*_rx register
4. The delay are applied for 6 pairs of CAN controller's TX and RX. The total round trip delay is dominated by external transceiver. The sampling point in each controller can be adjusted to sample correct bus value.

IO Pad	CAN Controller's IO
PIN_B_xx_can0_tx	can0_tx
PIN_B_xx_can1_tx	can1_tx
PIN_B_xx_enet1_txd_3	can2_tx
PIN_B_xx_enet1_mdc	can3_tx
PIN_B_xx_enet1_ptp_pps	can4_tx
PIN_B_xx_enet1_gtx_clk	can5_tx
PIN_B_xx_can0_rx	can0_rx
PIN_B_xx_can1_rx	can1_rx
PIN_B_xx_enet1_txen	can2_rx
PIN_B_xx_enet1_rxdv	can3_rx
PIN_B_xx_enet1_mdio	can4_rx
PIN_B_xx_enet1_clk_rx	can5_rx

Table 4-33. CAN Controller IOs and their Corresponding IOPADs.

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5. PACKAGE

The CV22FS55 chip has a 441-pin FC TFBGA package (14 mm x 14 mm).

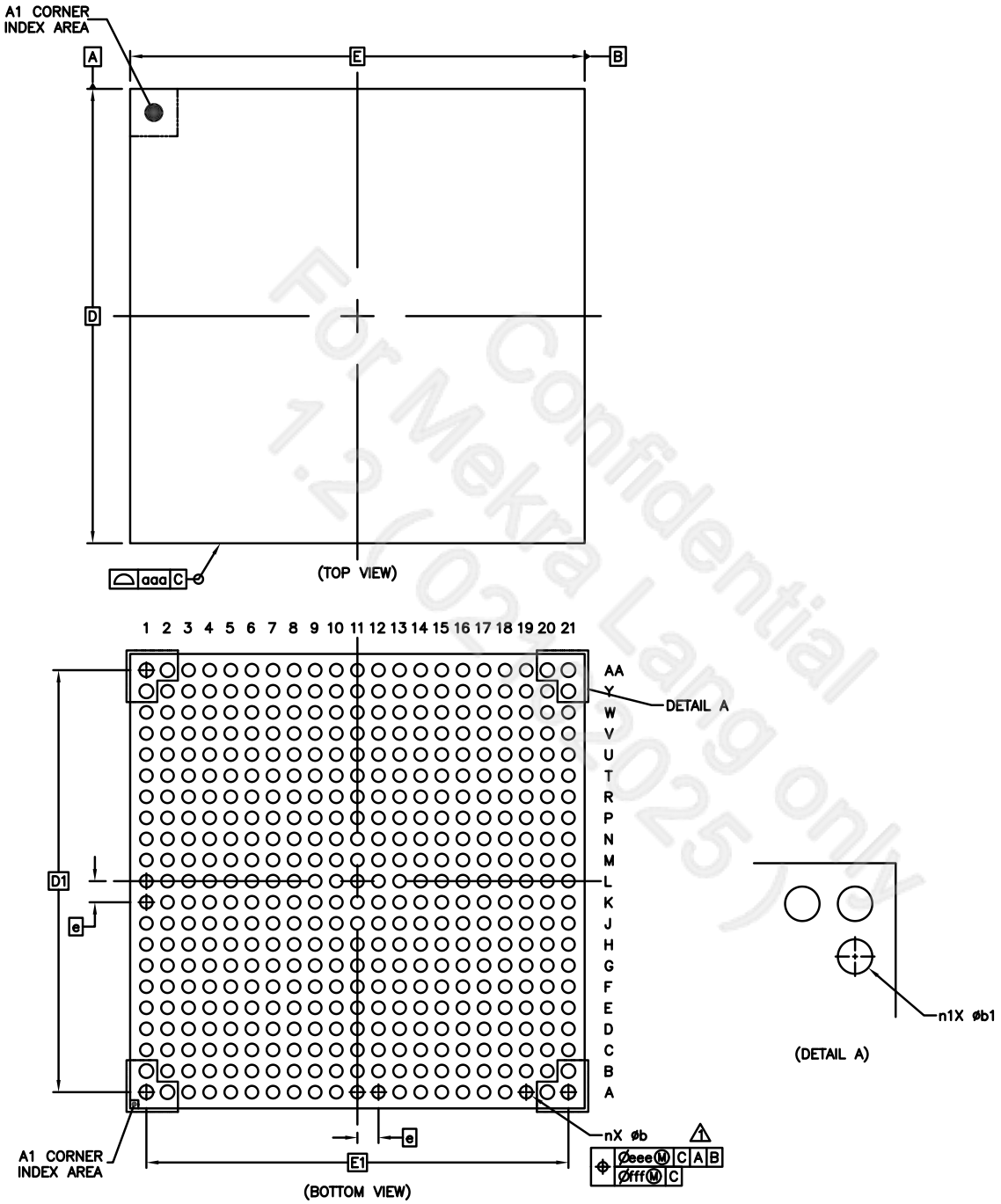


Figure 5-1. Top / Bottom View of the CV22FS55 Package with Detail A.

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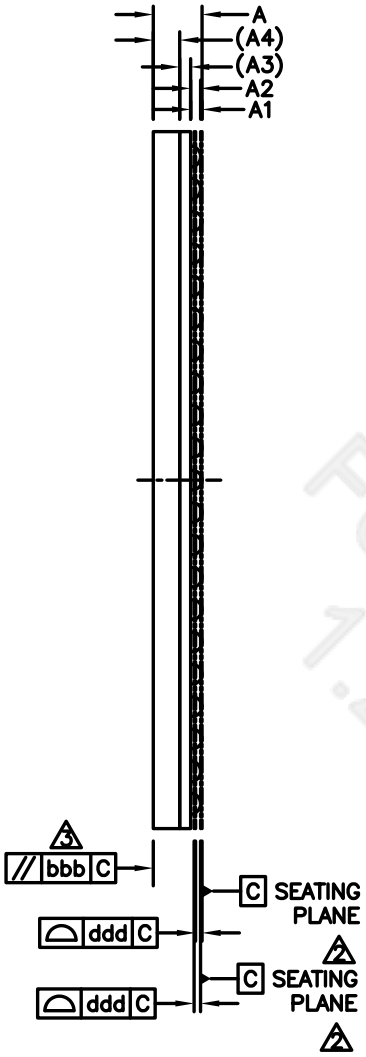


Figure 5-2. Side View of the CV22FS55 Package.

Description	Symbol	Dimensions		
		Minimum	Nominal	Maximum
Total Thickness	A	0.905	0.98	1.055
Stand Off	A1	0.18	0.23	0.28
	A2	0.15	0.2	0.25
Substrate Thickness	A3		0.22 REF	
Mold Thickness	A4		0.53 REF	
Body Size	D		14 BSC	
	E		14 BSC	
Ball Diameter			0.35	
Ball Opening			0.375 / 0.425	
Ball Width	b	0.34	0.39	0.44

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Description	Symbol	Dimensions		
		Minimum	Nominal	Maximum
	b1	0.375	0.425	0.475
Ball Pitch	e		0.65 BSC	
Ball Count	n		429	
	n1		12	
Edge Ball Center to Center	D1		13 BSC	
	E1		13 BSC	
Body Center to Contact Ball	SD		-	
	SE		-	
Package Edge Tolerance	aaa		0.1	
Mold Flatness	bbb		0.1	
Coplanarity	ddd		0.15	
Ball Offset (Package)	eee		0.15	
Ball Offset (Ball)	fff		0.08	
	Number of Balls: 441			

Table 5-1. Dimensions of the CV22FS55 Package (millimeters).

Notes:

NOTES:

1. DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO DATUM PLANE C.
2. DATUM C (SEATING PLANE) IS DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.
3. PARALLELISM MEASUREMENT SHALL EXCLUDE ANY EFFECT OF MARK ON TOP SURFACE OF PACKAGE.

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6. CONTACT AND ORDER INFORMATION

All chips in the CV22FS series are lead-free, halogen-free, and RoHS compliant.

Order Information: CV22FS55-A1-RH

Notes:

- CV22FS55: Chip Part Number
- A1: Version
- RH: In Memory of Rachel Hsu
- Engineering samples can be identified by the addition of an 'E' on the 2nd digit of the 3 digit XXX marking for Fab and Assembly Lot Number. Engineering samples are pre-qualification and pre-characterization parts, and are subject to the disclaimers governing the engineering sample (ES) of terms of service (ToS) agreement

For complete Ambarella contact information, please visit www.ambarella.com.

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NOTE: Page/chapter numbers for previous drafts may differ from those in the current version.

Version	Date	Comments
0.1	9 December 2019	Initial Version
0.2	17 January 2020	Updated Table 5-1 Dimensions of the CV22FS55 package
0.3	5 March 2020	Updated chip image and key features on title page Updated Section 2.3 VIN - updated LVDS and LVCMOS Updated Section 2.3 VIN - updated MIPI CSI-2 and SLVS port information Updated Section 1.2 Feature List - functional safety and added computer vision processing
0.4	10 June 2020	Updated Section 1.2 Feature List - functional safety - removed temperature sensors, updated SDXC SD card to maximum of 1 TB Updated Figure 4-2 Video Output Timing Updated Table 4-26 Timing Parameters for Serial NAND Updated Table 4-17 SD Timing Parameters and updated the note below the table Updated Section 1.2 Feature List to include sub-headings and added more details to some features including generic interrupt controller, RTC, and watchdog timers Updated Table 4-19 SD Controller Timing Parameters in SDR50 Mode Updated Figure 415 and Table 4-24 Timing parameters for SSI / SPI Master to delete t_{SDZ} Updated Table 4-5 DRAM I/O Characteristics - LVSTL I/O AC Specifications (Preliminary) Added Chapter 7 Errata Added Section 4.12 SPI NOR Timing Added Section 4.13 CAN Bus Timing Removed references to JPEG codec in Key Features and Figure 1-1 Functional Block Diagram Updated Table 4-18 Threshold Level for High Voltage to max of 3.3 V Updated Table 4-20 Threshold Level for 1.8 V Signaling to max of 3.3 V Updated Table 4-3 Digital I/O Characteristics to max of 3.3 V Updated Table 4-2 Power Rails DC Characteristics to max of 3.3 V Added Crystal Clock max frequency tolerance Updated R52 Clock frequency to be 456 MHz Updated encoding to 240 MPixel/s (8MP30) Updated H.265 level information to Section 1.2 Feature List

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Version	Date	Comments
0.5	1 December 2020	Updated Section 4.13 CAN Bus Timing--deleted timing diagram, updated table and notes Updated Table 4-28 Timing Parameters for SPI NOR and added some notes Added a note to Table 4-26 Timing Parameters for Serial NAND Updated package type to FC TFBGA Updated Figure 4-8 SD Host Output Timing Updated Section 4.8 I2C Timing Updated Section 4.9 SSI / SPI Timing Updated SMIO_29 pin location in Table 3-10 Storage Media Interface Pins (SMIO) in NAND Flash and SD Modes Updated Figure 1-1 Functional Block Diagram Updated Section 1.2 Features List—Advanced Security Features Updated Section 2.16 JTAG to delete note Updated Section 1.2.9 Video encoding Updated Chapter 5 Package and Chapter 6 Contact and Order Information for Rev A1 package Deleted Chapter 7 Errata as the issue is resolved in Rev A1
0.6	1 March 2021	Updated Table 4-32 Timing Parameters for Serial NAND to update names of configuration registers Updated Section 3.2.3 Pins: Sensor/Video Input to add MIPI CSI-2 pin note Updated fast mode minimum value for Fall Time in Table 4-33: I2C Characteristic of SDA and SCL for F/S-Mode Removed VIL-ULP from Table 4-8 MIPI LP Receiver DC Specification Deleted Table 4-18 Threshold Level for High Voltage Deleted Table 4-19 Threshold Level for 1.8V Signalling Updated CAN TX RX minimum values in Table 4-33 Timing Parameters for CAN Bus Updated Figure 1-1 Functional Block Diagram to remove SLVS and LVCMOS Updated Section 1.2.6 Sensor/VIN Interfaces to remove SLVS, LVCMOS, and LVDS Updated Section 2.3 Interfaces: VIN to remove SLVS, LVCMOS, and LVDS Updated Section 3.2.2 Pins: Sensor/VIN to remove LVDS, SLVS, and LVCMOS Deleted Section 4.3.5.1 VIN: SLVS/LVCMOS IO Updated Table 4-11 caption to remove SLVS
0.7	2 July 2021	Updated Figure 4-9 Clock Signal Timing Updated Section 1.2.16 Advanced Security Features Updated Table 4-1 Absolute Ratings Updated Table 5-1 Dimensions of CV22FS55 Package Updated Section 2.4.2 MIPI CSI-2
0.8	6 October 2021	Updated Section 2.4.1 MIPI DSI Updated Table 4-29 Timing Parameters for Serial NAND and added a note Updated title page and Section 1.2.5 Functional Safety to update processing island to be targeted to meet ASIL C requirements
0.9	5 April 2022	Updated title page and Section 1.2.5 Functional Safety
1.0	9 August 2023	Updated LPDDR4 max capacity to 2 GB from 4 GB Added a note to the Chapter 6: Contact and Order Information Updated Section 4.12 SPI NOR timing Updated Section 2.8: SD / SDIO Updated Sections 3.2.6 Gigabit Ethernet Interface and 3.2.9 SMIO Interface Updated Section 2.14: RTC

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Version	Date	Comments
1.1	4 January 2024	Updated Figure 4-27 Timing for Serial NAND Hold and Figure 4-10 SD Host Input Timing Added a note after Table 4-3 Digital I/O Characteristics
1.2	29 January 2025	Updated Table 3-2 DRAM Pins

Table 8-1. Revision History.

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